

ELITE



OPHTHALMOLOGY

A compiled note for Medical PG Entrance



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More than just notes...

ELITE OPHTHALMOLOGY

A compilation note for Medical PG Entrance Exam

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OPHTHALMOLOGY

notes

We hope you enjoy using these notes

They have been hand-crafted with an obsessive attention to details, with aim of capturing the course's content in a way that's right for you.

Thank You

*Inspired by
Samuel Suresh*



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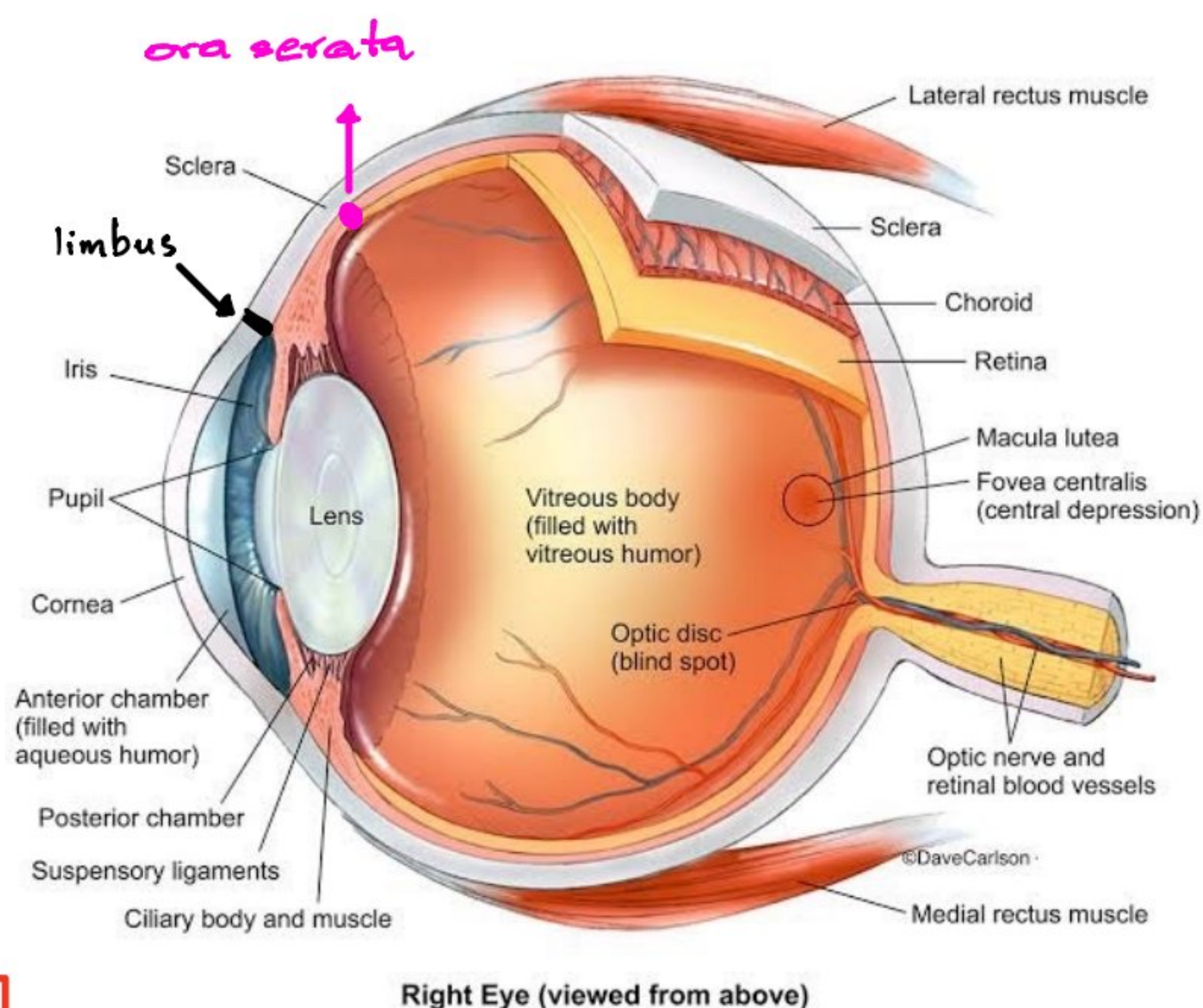
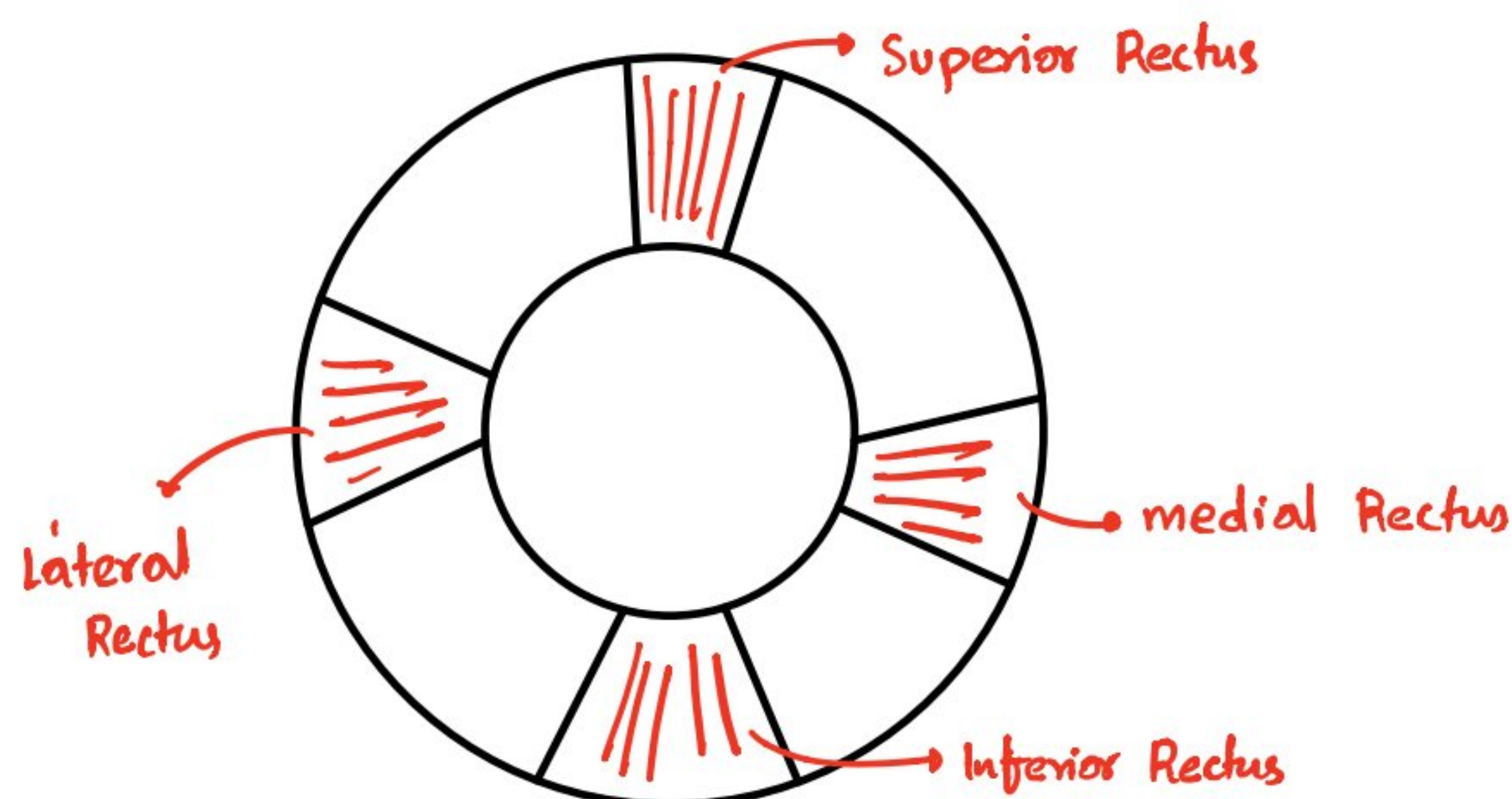
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Reference:

- ***K.D Tripathi Textbooks***
- ***Shanbag Textbook***
- ***Khurana Textbook***
- ***Dr. Rajaratna Lecture Class***
- ***Dr. Sourabh Lecture Class***

ANATOMY OF EYE:

- Volume of orbital cavity = 30ml
- Eyeball / Globe $\Rightarrow$ Volume = 6ml



7 attachment = 6 muscle + 1 optic nerve
Removal of attachment = Enucleation

7 attachment $\rightarrow$ M.R + S.O + Optic Nerve
L.R I.O
S.R
I.R
 $\Rightarrow$ six EOM

The shape of the eyeball is a sphere
Volume = 6ml
Weight = 7gm
Circumference = 7.5cm

Layers of Eyeball

1) Outer fibrous layer $\rightarrow$ Cornea { Ant. $\frac{1}{6}^{th}$ }
 $\rightarrow$ Sclera { Post. $\frac{5}{6}^{th}$ } } They are thick & strong to sustain the IOP

• Junction b/w cornea & sclera $\rightarrow$ Limbus

2) Middle vascular layer $\rightarrow$ Uveal layer

Iris $\rightarrow$ ciliary body $\rightarrow$ Choroid

(Ant.) $\rightarrow$ (Post.)

• Junction b/w ciliary body and Choroid $\rightarrow$ ORA SERATA

$\approx$ B/w Pars Plana & Choroid

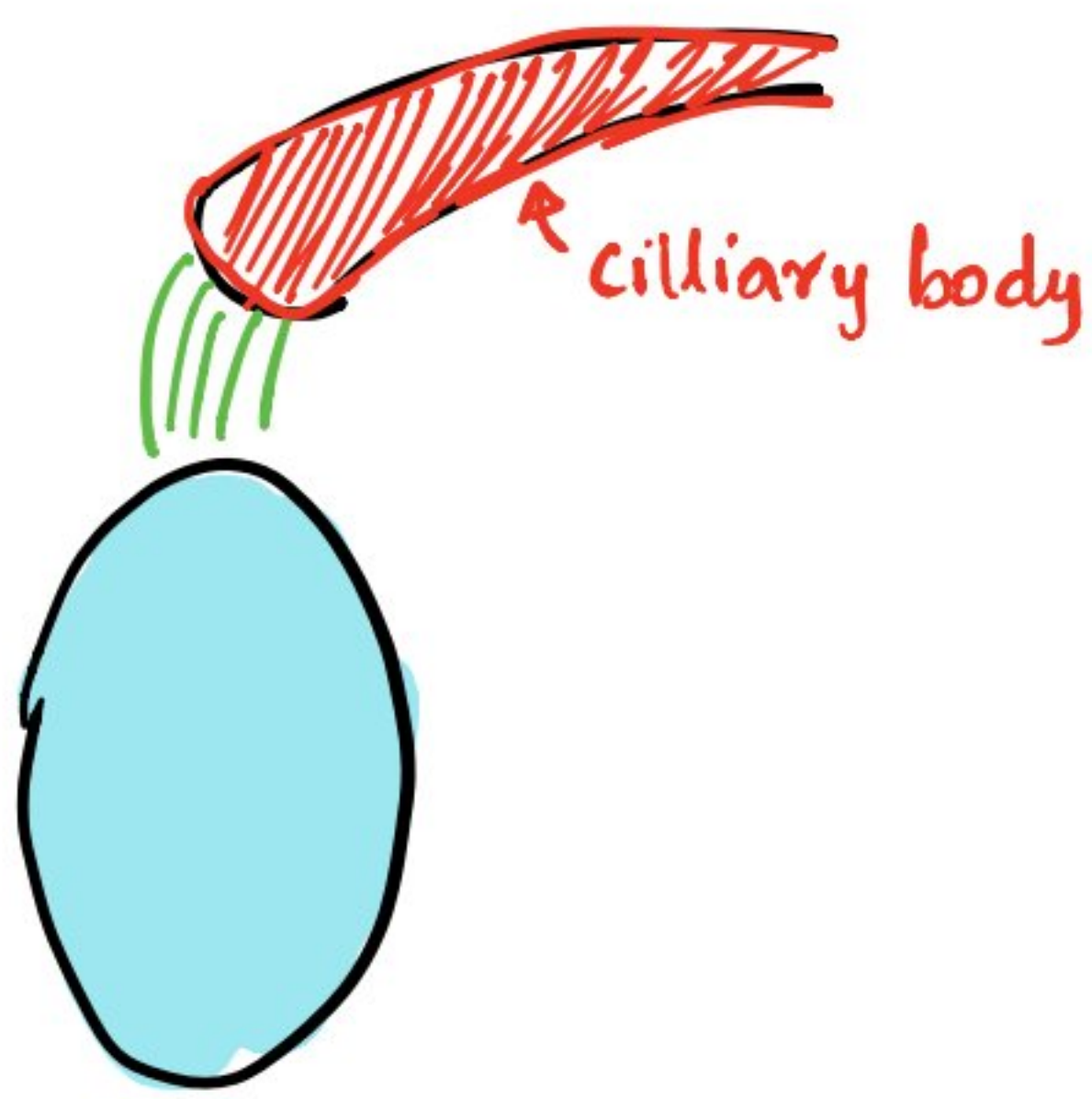
AIMS/JIPMER

60-70 ciliary process

(A) Pars Plicata

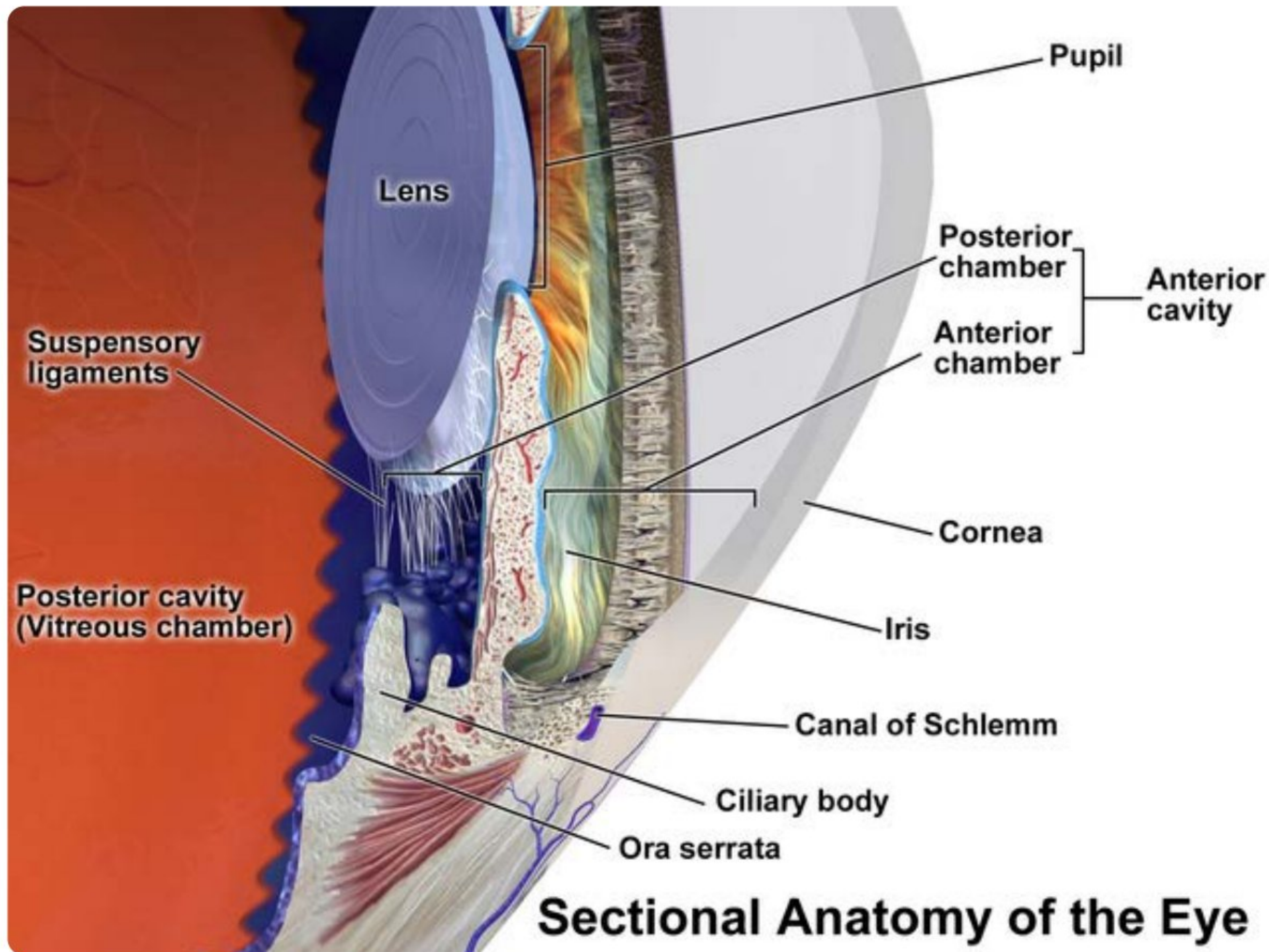
(P) Pars Plana

• Point of attachment of ciliary body to sclera $\rightarrow$ scleral spur



3) Inner most layer → neural layer → **Retina**.

From Ora Serrata → to → Optic disc



Segment of the Eyeball:

Anterior Segment → Anterior to posterior part of lens

Ant. Chamber

Post. Chamber

From Post. part of cornea till iris, lens & diaphragm

Iris to post. part of lens

Q: Lens situated in the posterior chamber of the anterior segment

Content ⇒ Aqueous Humour

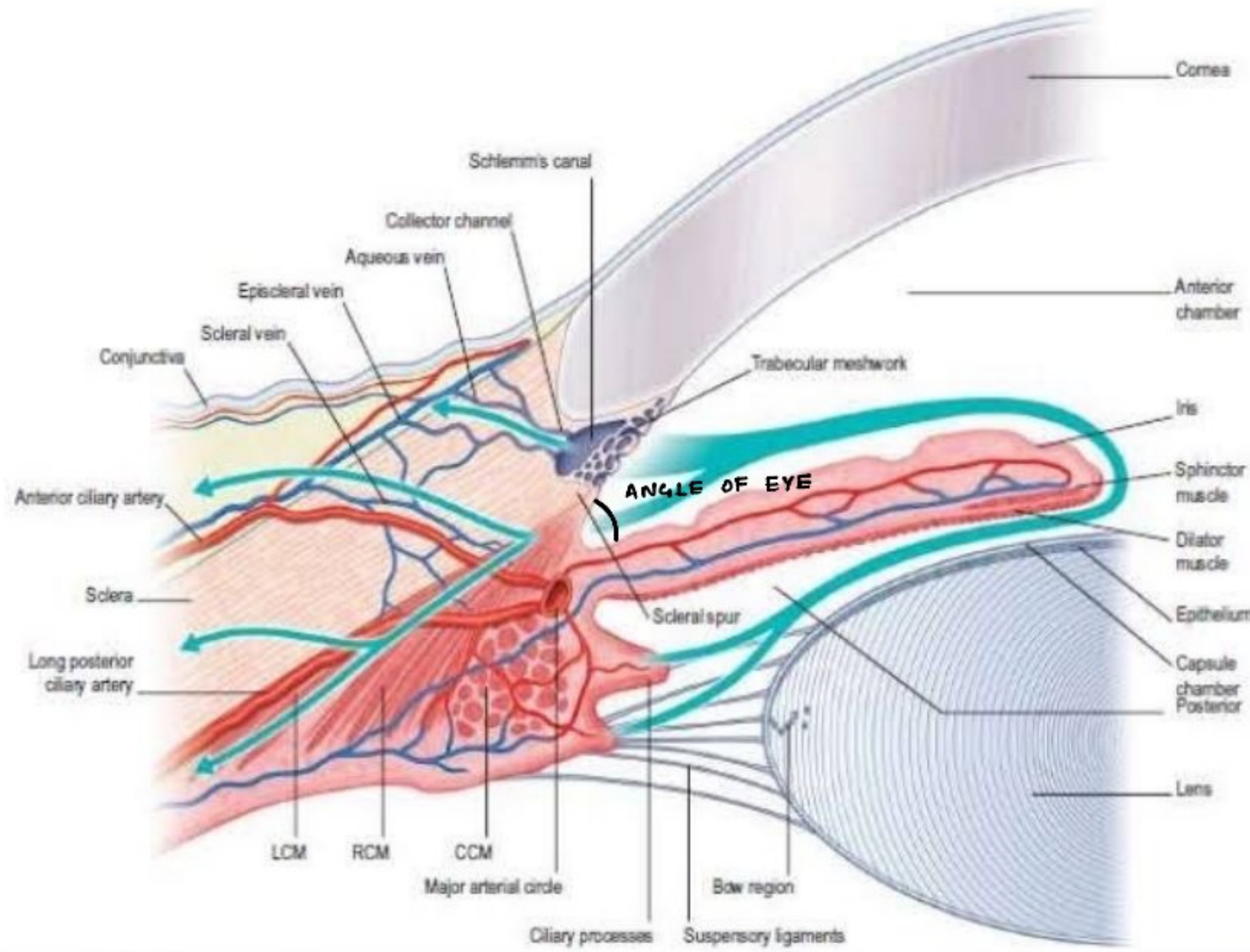
AC = 0.25ml	} 0.3ml
PC = 0.05ml	

• Filtrate of plasma formed from the ciliary process {60-80}

• Rate of formation = **2 - 2.5ml/min**

- Formed by 3 process →
 - i) Secretion ⇒ Maxm. Active process, **ATP required** (70%)
 - ii) Diffusion ⇒ 20% passive process, **No ATP required**
 - iii) Ultrafiltration ⇒ 10% passive process, **No ATP required**

⚡ Aqueous humour formed in the posterior chamber of the anterior segment



Three way of drainage

① Trabecular Outflow = 90%

⚡ {0.22 - 0.28 ml / min / mmHg}

post. chamber → pupil → Trabecular meshwork

Aqueous / conjunctival vein

Schlemm's Canal

EPISCLERAL VEIN

Type of Trabecular meshwork:

- i) Corneoscleral
- ii) Uveal
- iii) Juxta canalicular

⚡ Maximum resistance to flow of aqueous → at Juxta canalicular trabecular meshwork (close to Schlemm's canal)

To release resistance in Glaucoma → Trabeculectomy

② Uveoscleral outflow = 10%

drain → Post. Ciliary vessels

Retina
Choroid
supra choroid space
Sclera

③ Uveovortex outflow = minimal %

drain → Vortex Vein

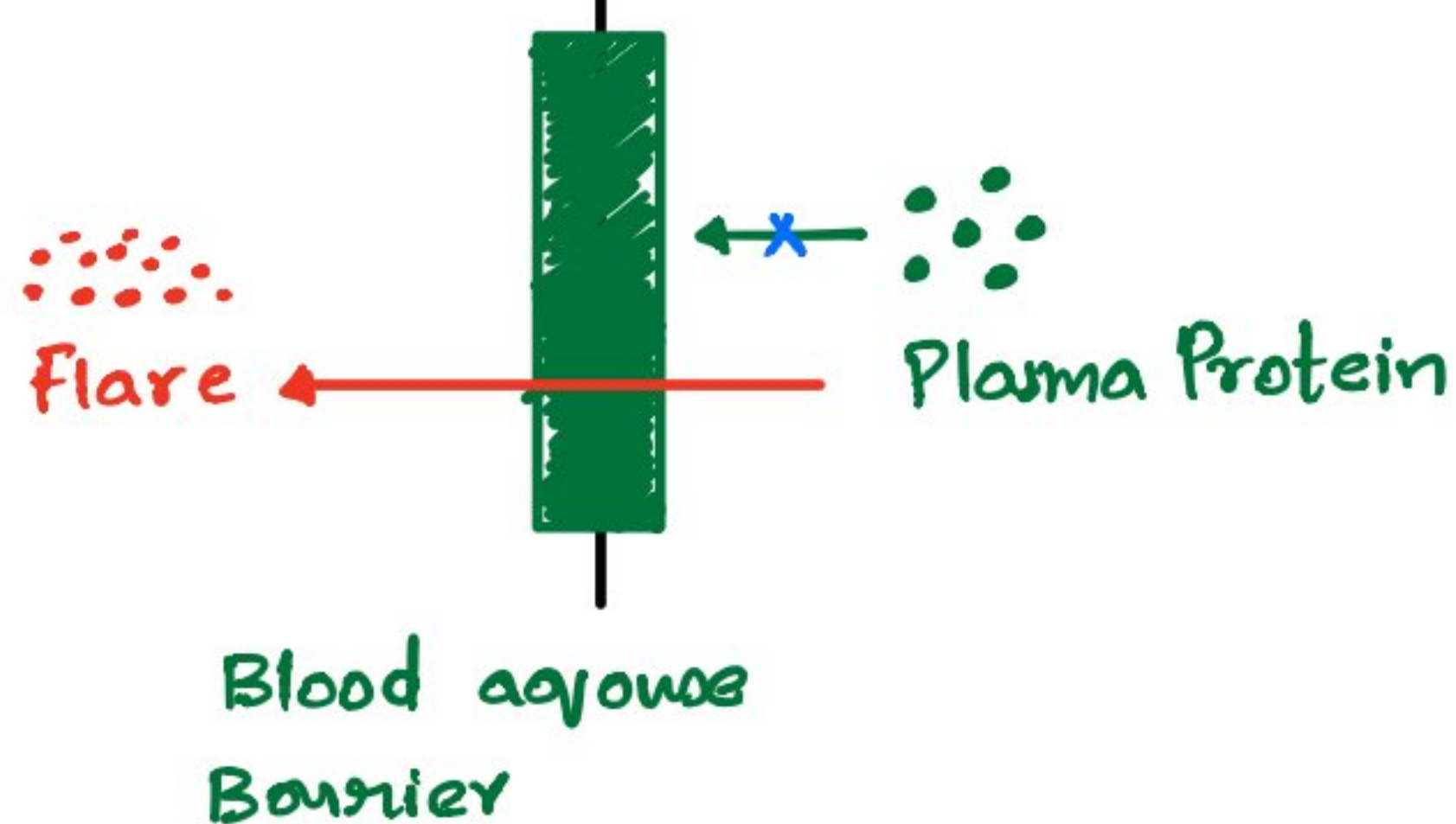
Posterior segment → Post to posterior part of lens

Content → Vitreous Humour
Volume = 4ml

ACQUOES	PLASMA	VITREOUSE
1) 99% water 2) Lactate, Ascorbate & Chlorine more in aqueous		1) 99% water 2) Collagen, Hyaluronic acid & Ascorbate are more in vitreous

than plasma

- Else is less compared to plasma



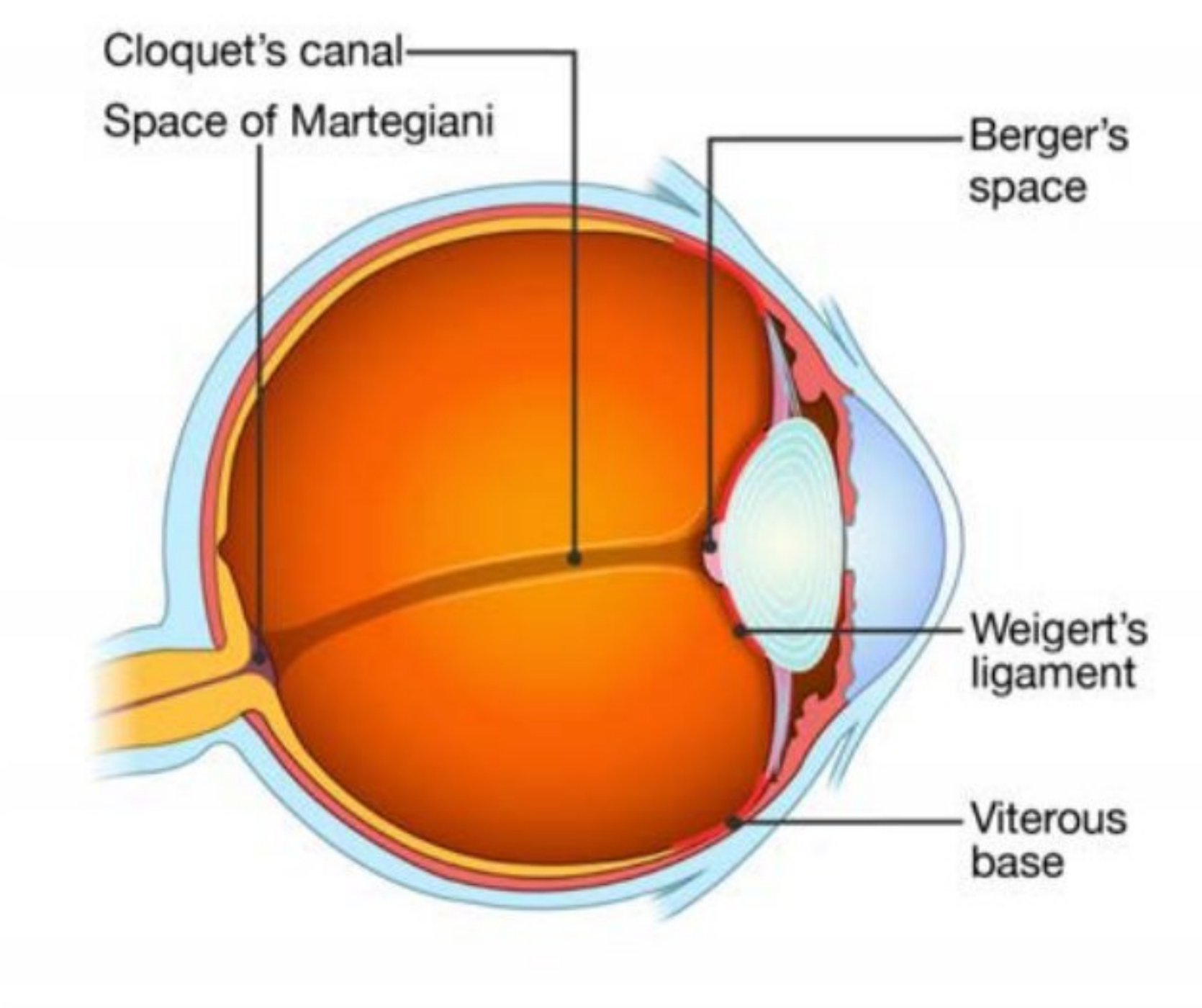
§ formed by tight junction of iris endothelium & ciliary non pigmented epithelium §
fx: prevent plasma protein in entering aqueous.

Pathology: Uveitis → Inflammation in Eye

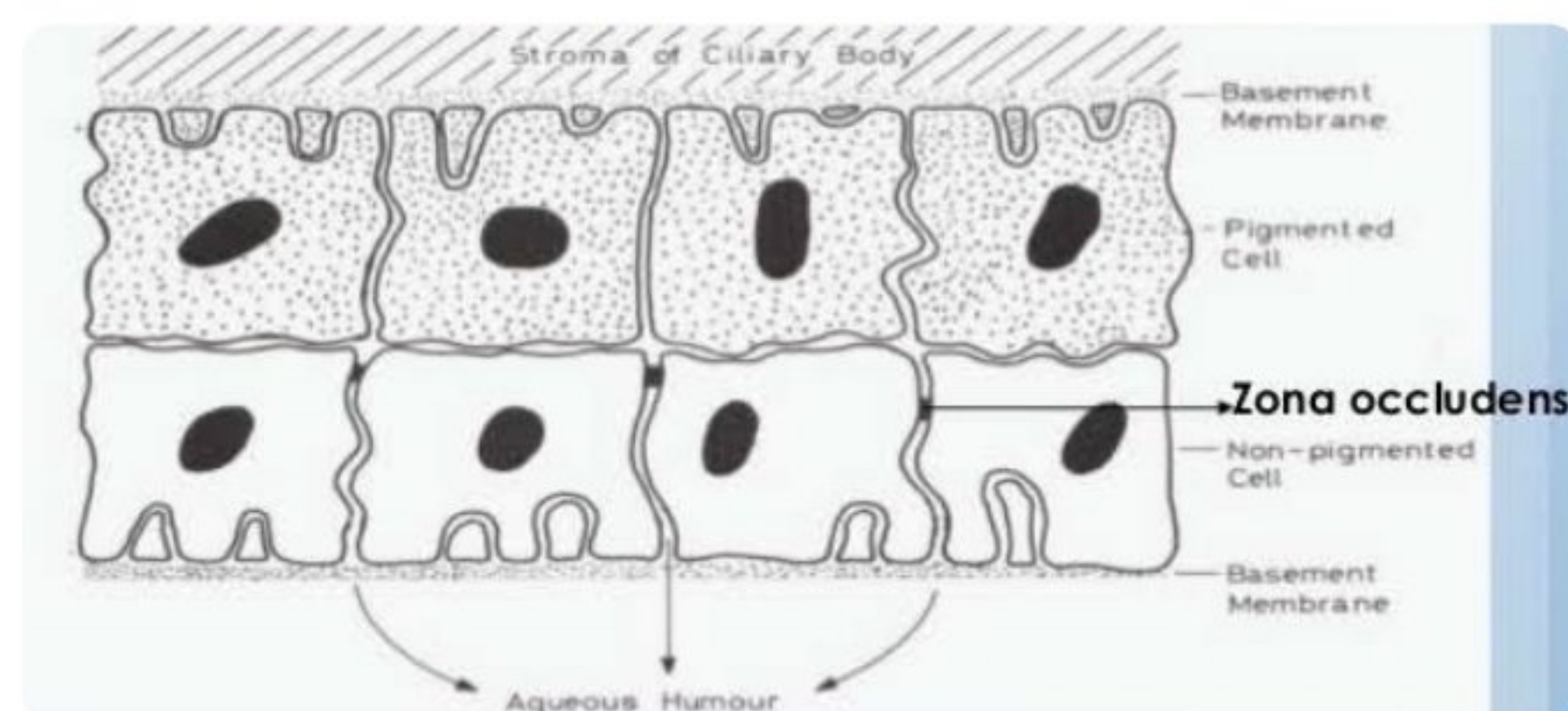
↓
Blood aqueous barrier broken down

↓
FLARE { Leakage of proteins in Aqueous

3, fluid like



compared to plasma
• else is less compared to plasma.

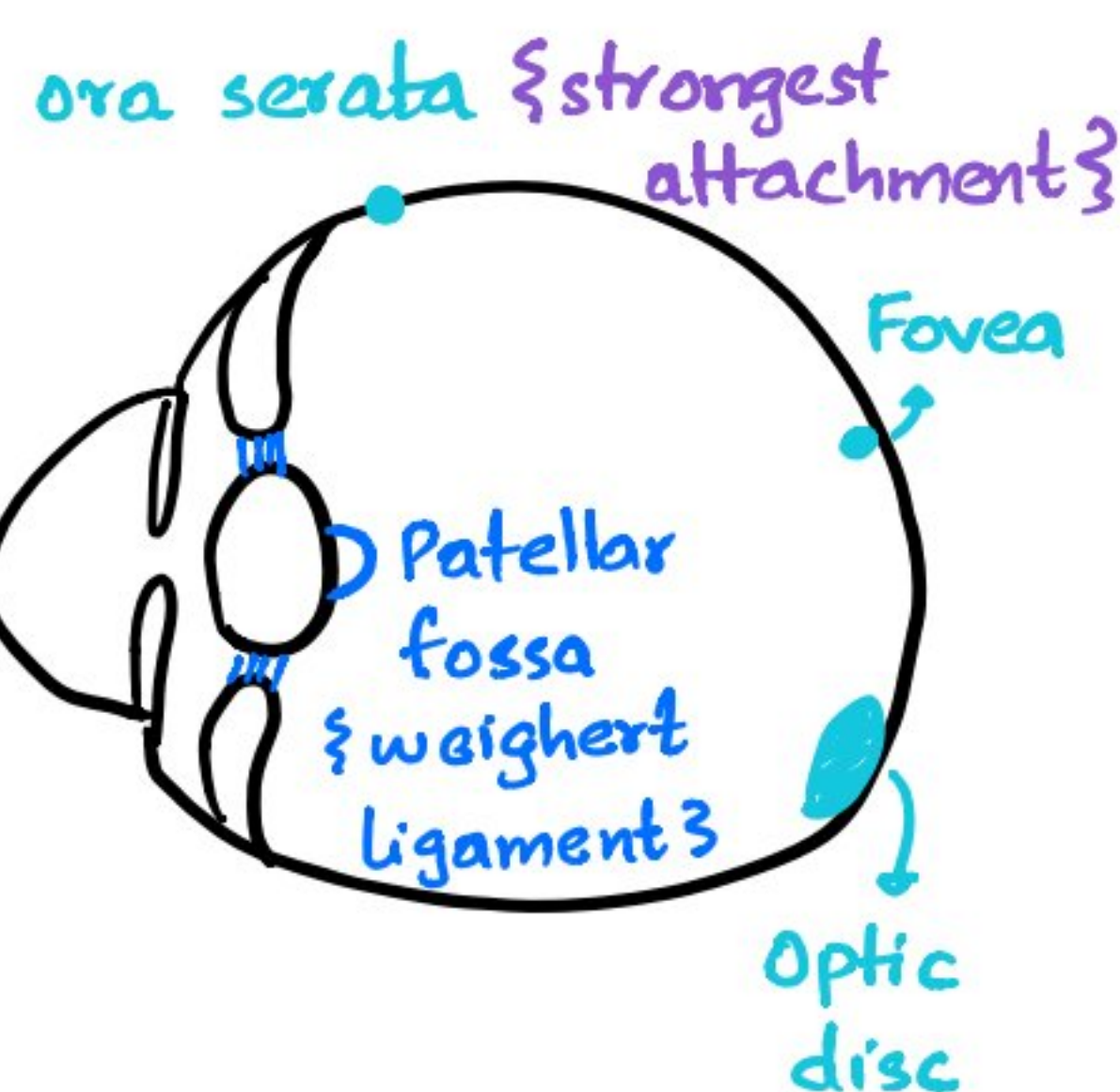


During Inflammation, the blood barrier gets broken down

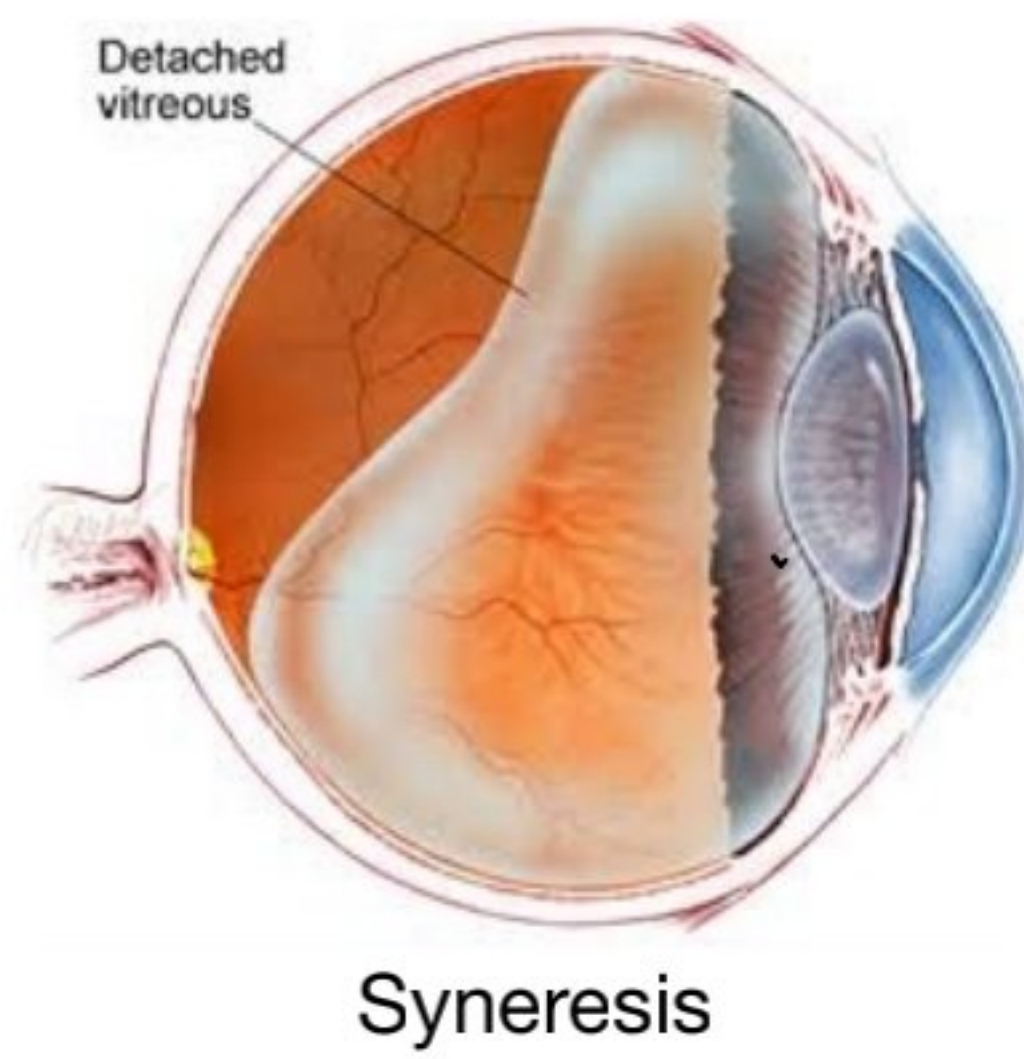
3, Gel like → d/t collagen
COLLAGEN II (MC)

VI
IX
XI

• Attachment of Vitreous



Weighert ligament → surround the patellar fossa
{ strong in children }



• When we age $\Rightarrow$ gel vitreous collapse

K/a **SYNERESIS**

- Posterior vitreous collapse first } post. vitreous
- detachment from retina } may cause Retinal tear or hole

Liquified vitreous
SYNCHISIS

Enters Retinal Hole

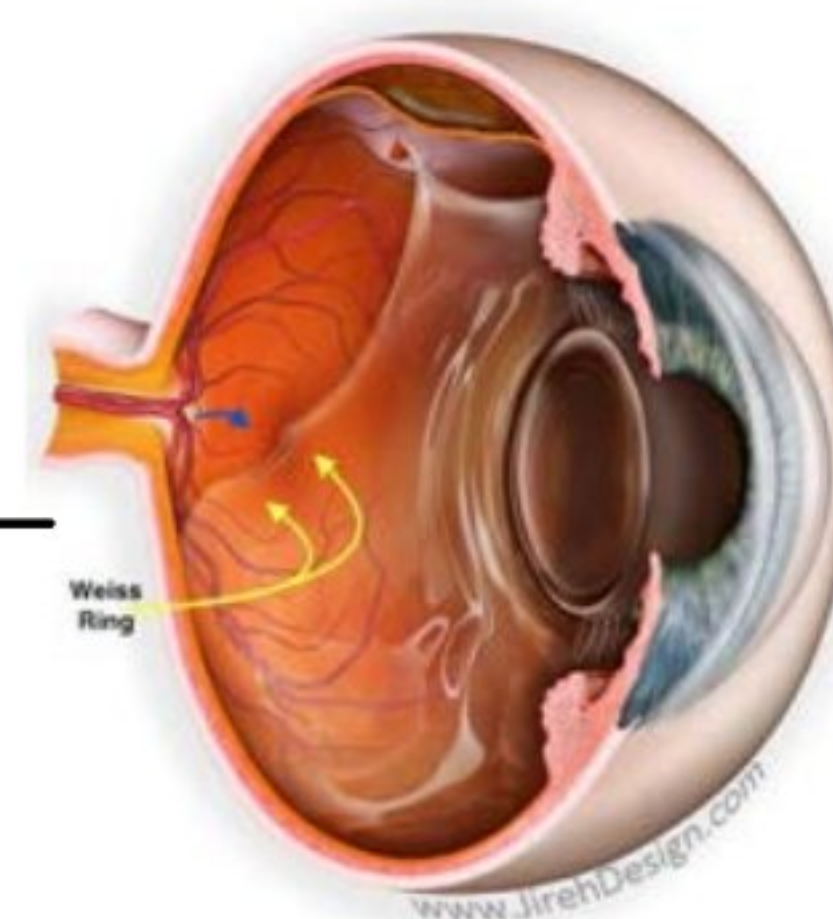
RETINAL DETACHMENT
& Rhegmatogenous

• Entire vitreous collapsing

seen as a Ring

WEISS RING

s/o Complete PVD



Symptoms :-

1) Scotoma $\rightarrow$ floater & visual field defect

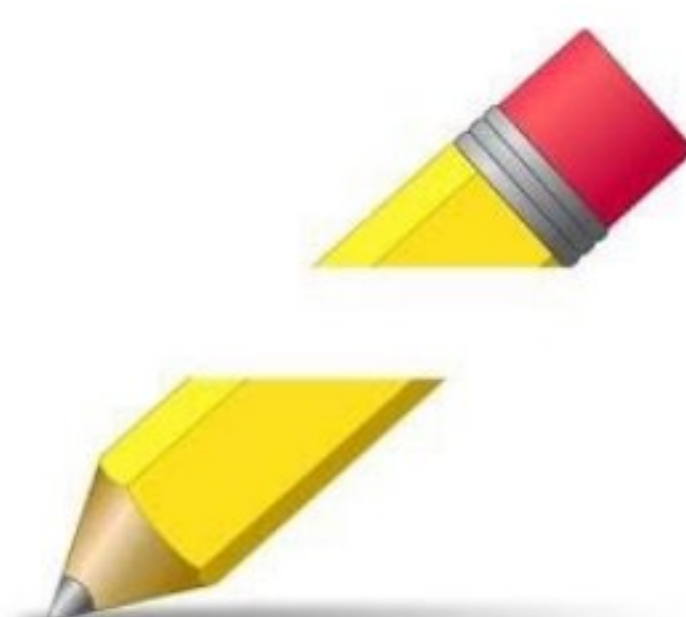
- Single
- Old age
- Positive scotoma

+ Negative Scotoma
Eg: Blind Spot

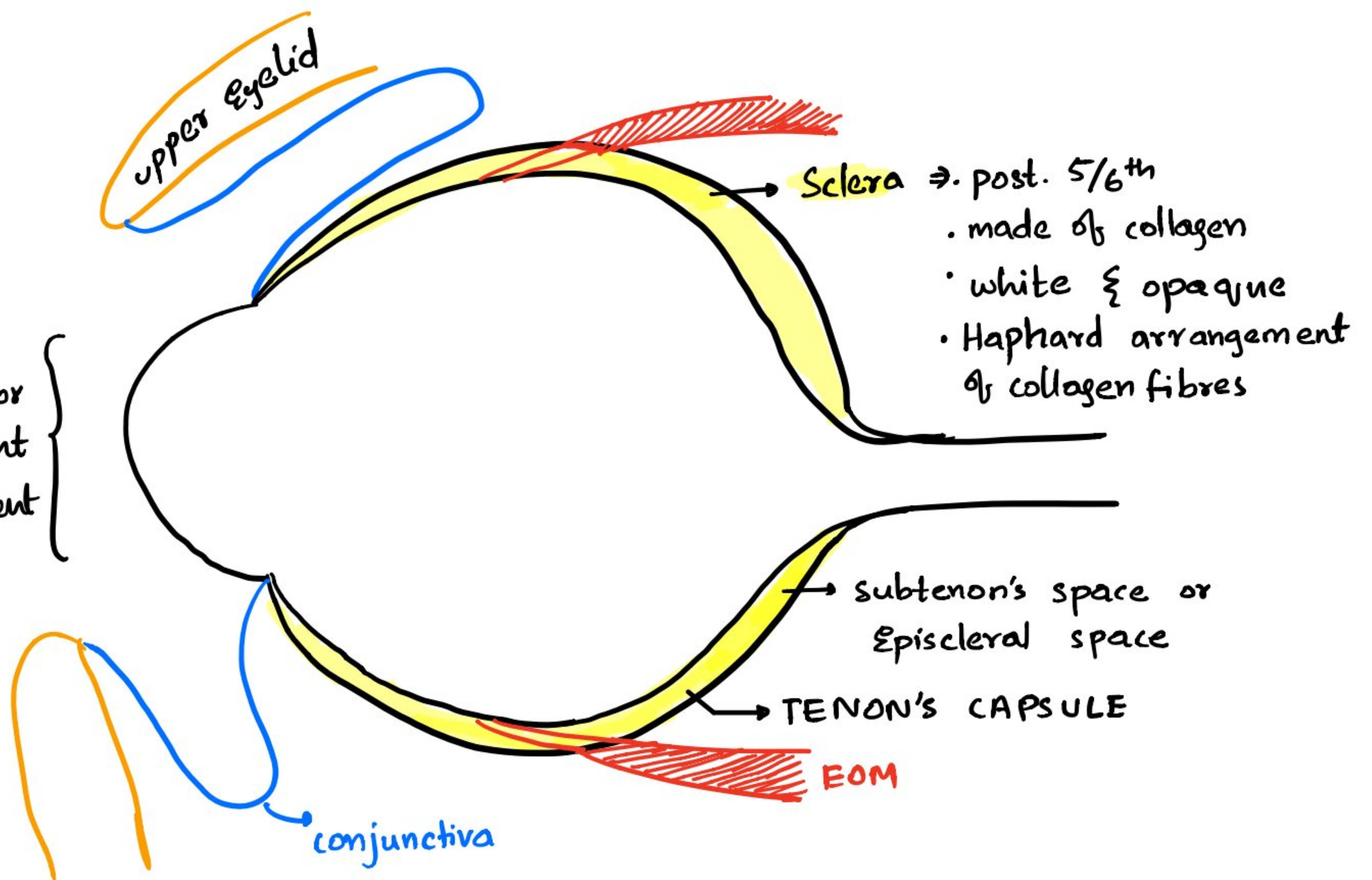
Scotoma

Negative Scotoma

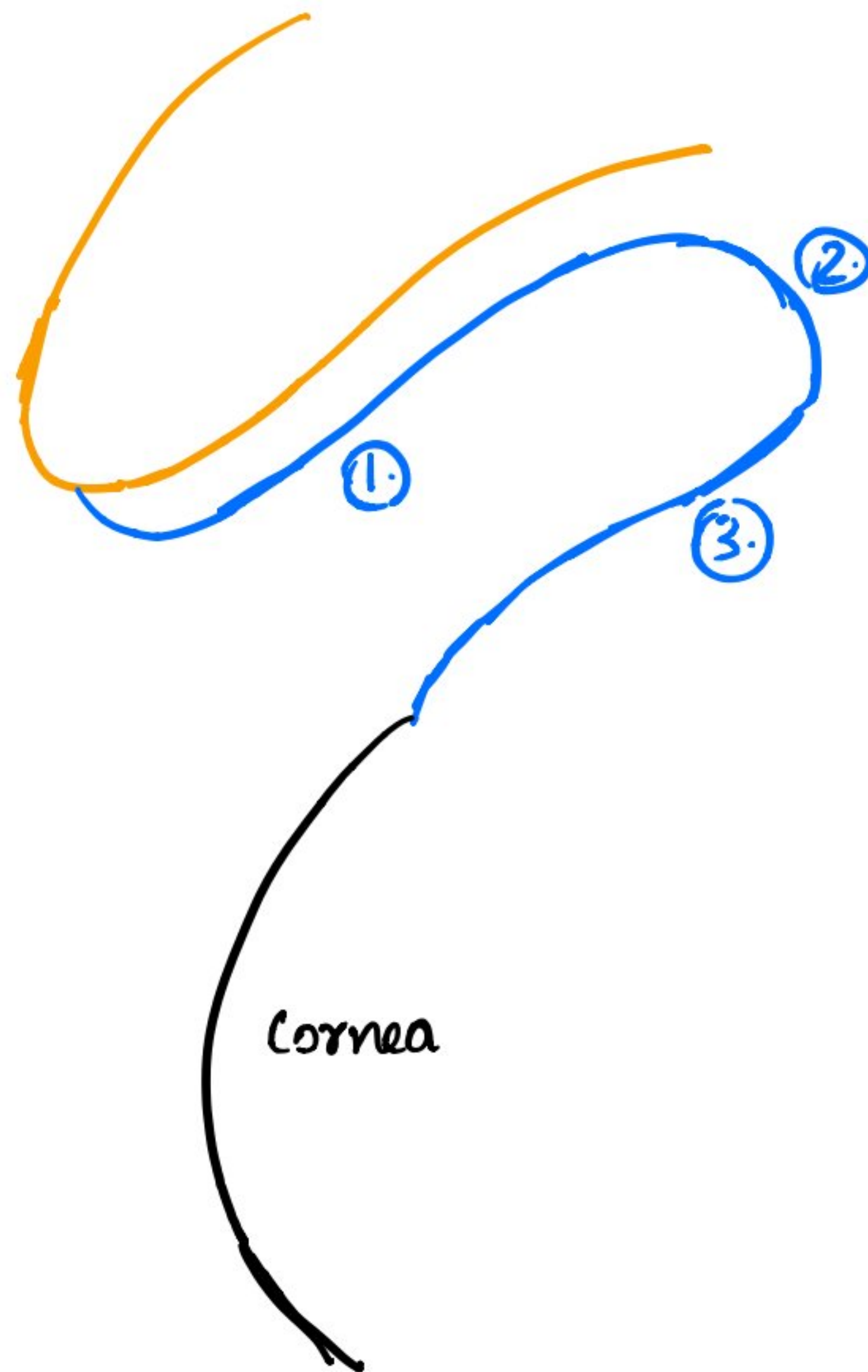
Positive Scotoma



Cornea $\Rightarrow$ $\frac{1}{6}^{\text{th}}$ Anterior
 • Transparent
 Regular arrangement
 of collagen fibres



Anatomy of Conjunctiva:



Parts of Conjunctiva

- ① Palpebral Conjunctiva
- ② Forniceal Conjunctiva
- ③ Bulbar Conjunctiva

Extra Ocular Muscles:

No. : Six

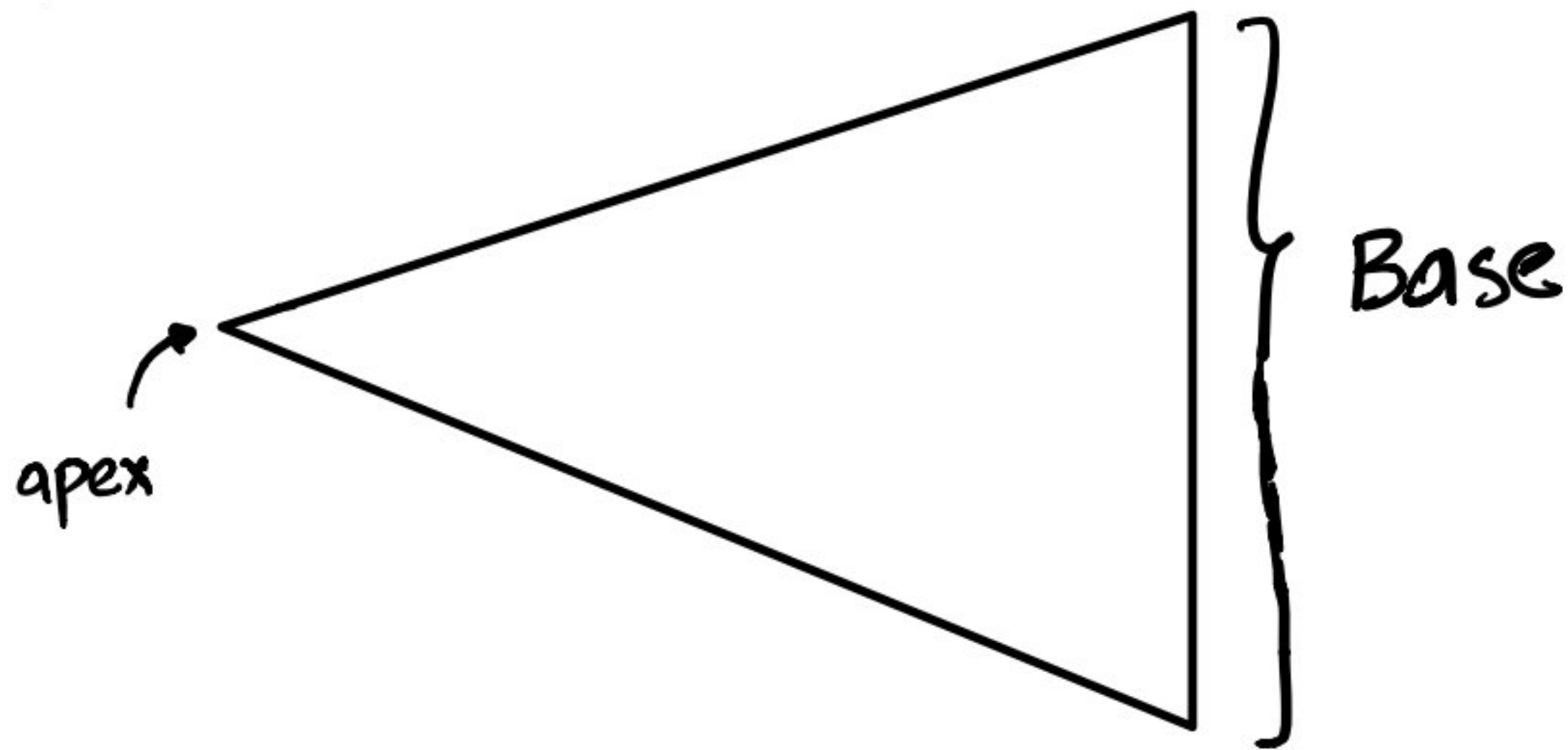
Origin : Apex of orbit

except $\rightarrow$ Inferior Oblique

Orbit:

Shape : Pyramid

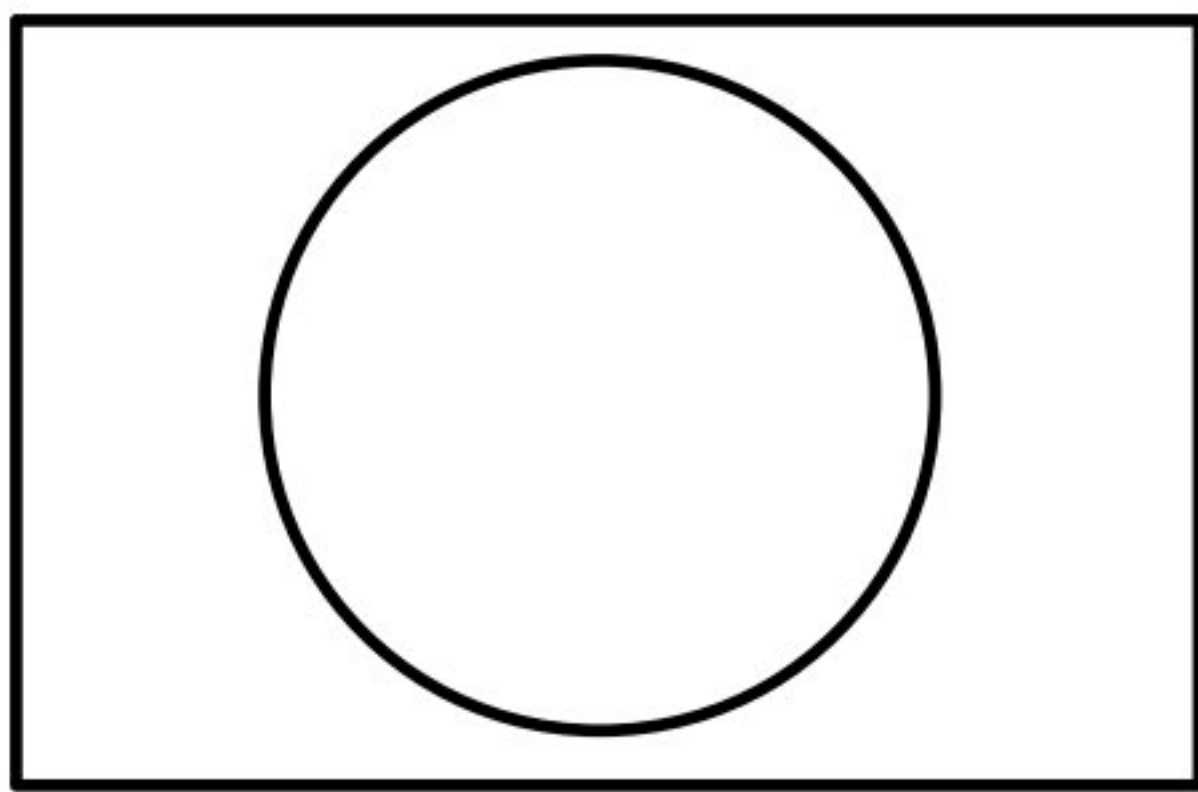
Volume : 30ml



Lesser Wing of sphenoid

Frontal

Zygomatic
Greater wing
of sphenoid



Body of Sphenoid

medial

(L)

(E)

Ethmoid

Lacrimal

PA

palatine

ZA

zygomatic

MA

Maxillary

- 7 bones present in the orbit
- Medial wall thinnest → Lamina Papyracea (thinnest bone ⇒ Ethmoidal bone)
Children : Ethmoidal sinusitis → MCC of orbital cellulitis in children
- Weakest wall / (MC) wall to # → Floor of orbit d/t infraorbital groove.
- Lateral wall strongest, thickest
- Weakest point of orbit → Posteromedial aspect of maxillary bone in floor
- Blunt trauma → floor #
or Blow out #

HYP0-ESTHESIA or ANAESTHESIA OF IPSILATERAL CHEEK

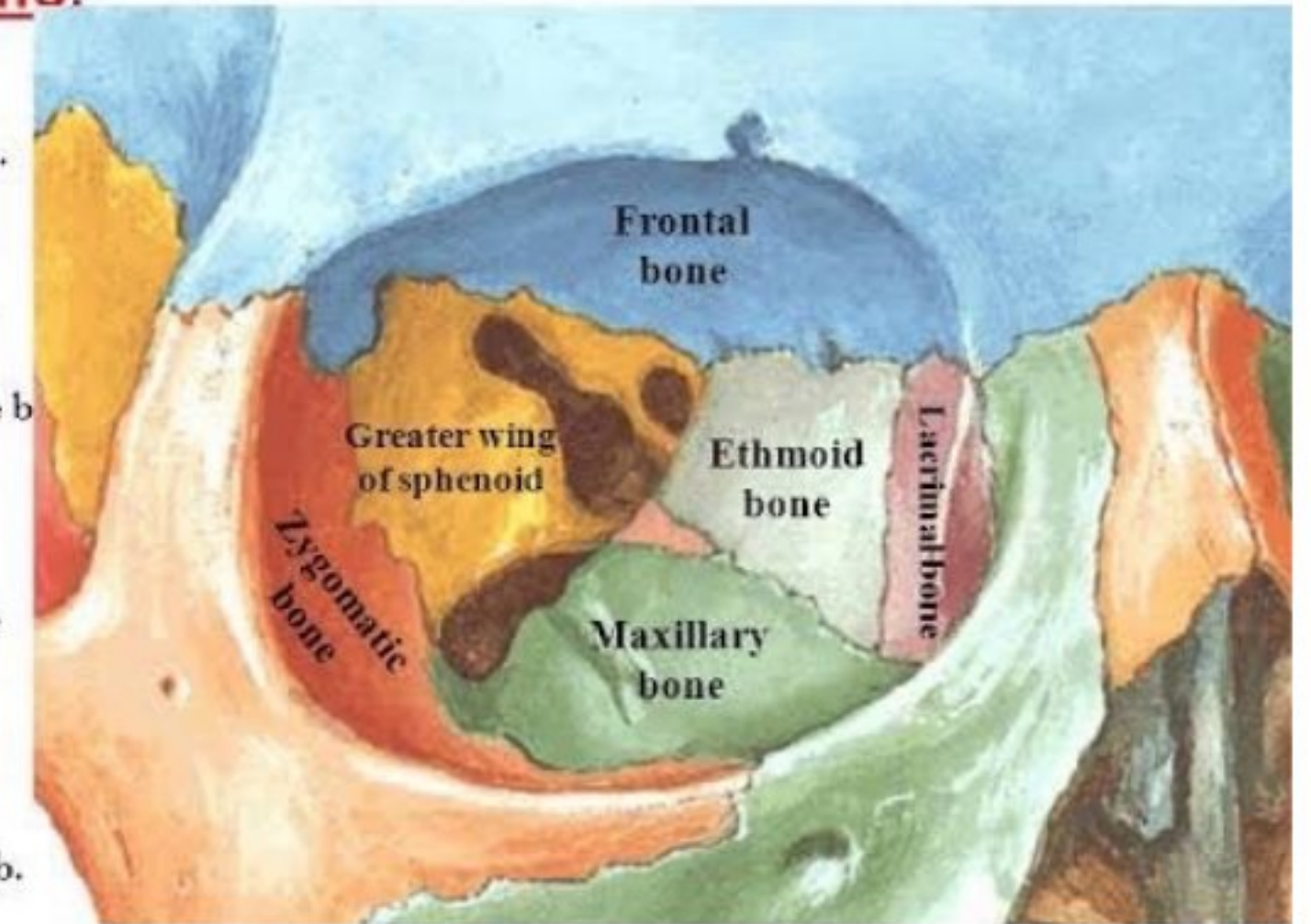
- Xray Sign ⇒ Tear drop sign (waters view)
- enophthalmos
- Inferior rectus can get entrapped in this #
↳ Difficulty to look up

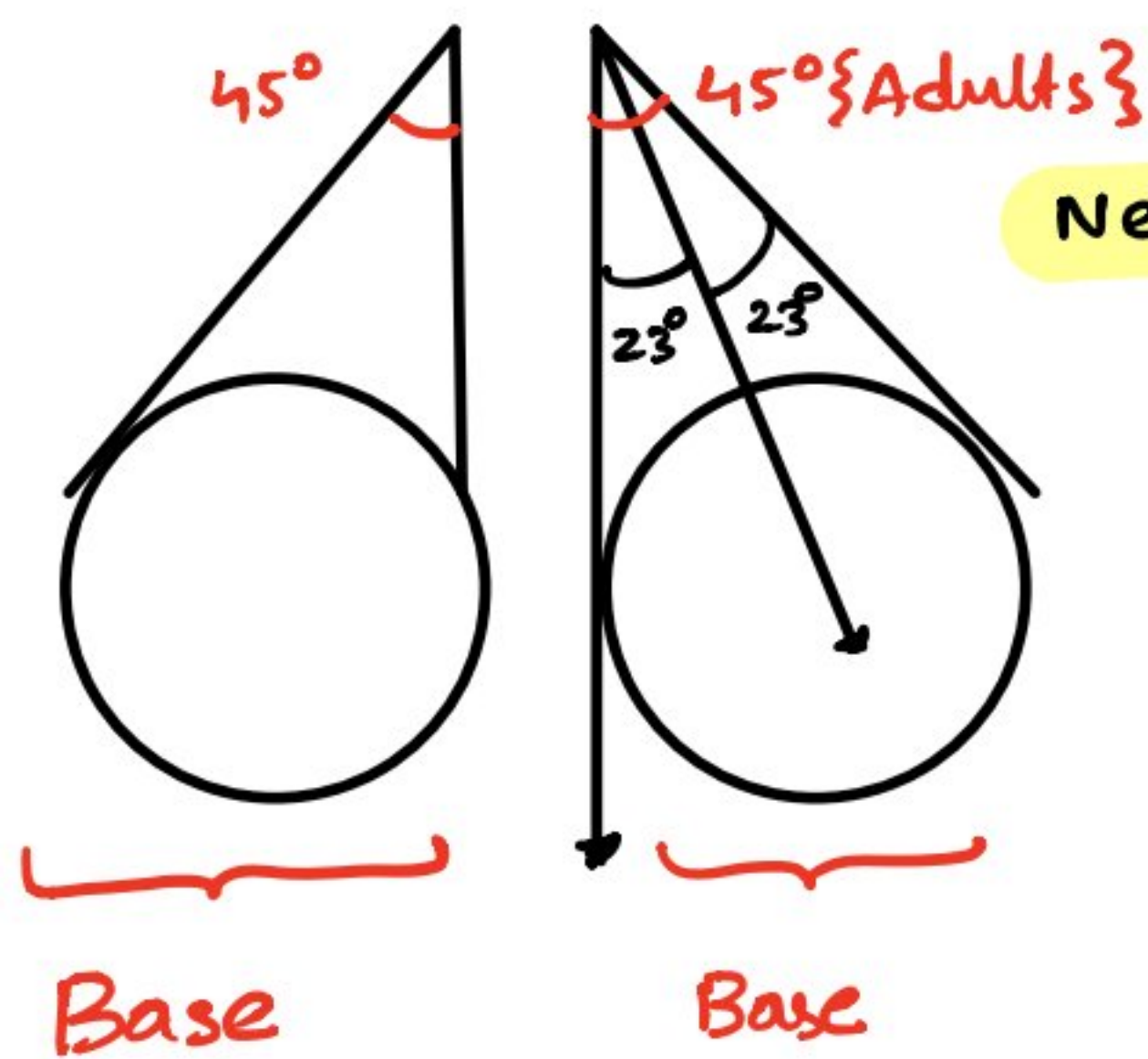
ORBITAL CAVITY

A pyramidal space with a base, apex and four walls.

Boundaries and Walls:

- **Roof**
 1. Orbital plate of frontal b.
 2. Lesser wing of sphenoid b.
- **Floor**
 1. Orbital surface of maxilla
 2. Zygomatic bone
 3. Orbital process of palatine b.
- **Medial wall**
 1. Lacrimal bone
 2. Orbital plate of ethmoid b.
 3. Body of sphenoid b.
- **Lateral wall**
 1. Zygomatic b.
 2. Greater wing of sphenoid b.





Eyeball & orbit are at 23°

- 3 gap b/w orbit and brain $\rightarrow$
- ① **Optic Canal** { optic N. and Ophthalmic artery }
gap of lesser wing of sphenoid
 - ② **Superior orbital fissure**
gap b/w lesser and greater wing of sphenoid
 - ③ **Inferior orbital fissure**
Maxillary Bone & Zygomatic

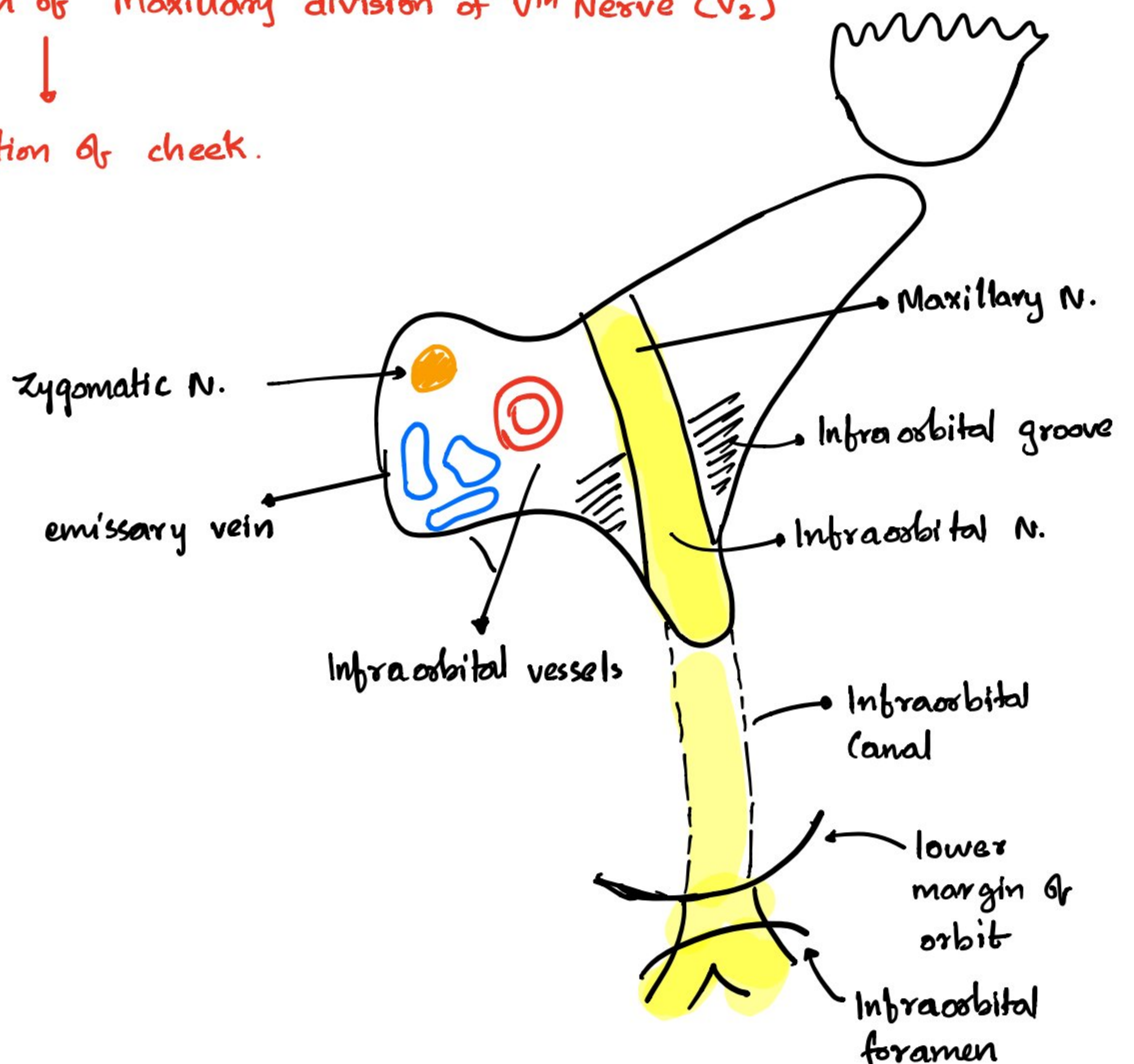
Intra Orbital fissure $\rightarrow$ @ floor of orbit
 $\rightarrow$ contains the infraorbital N.



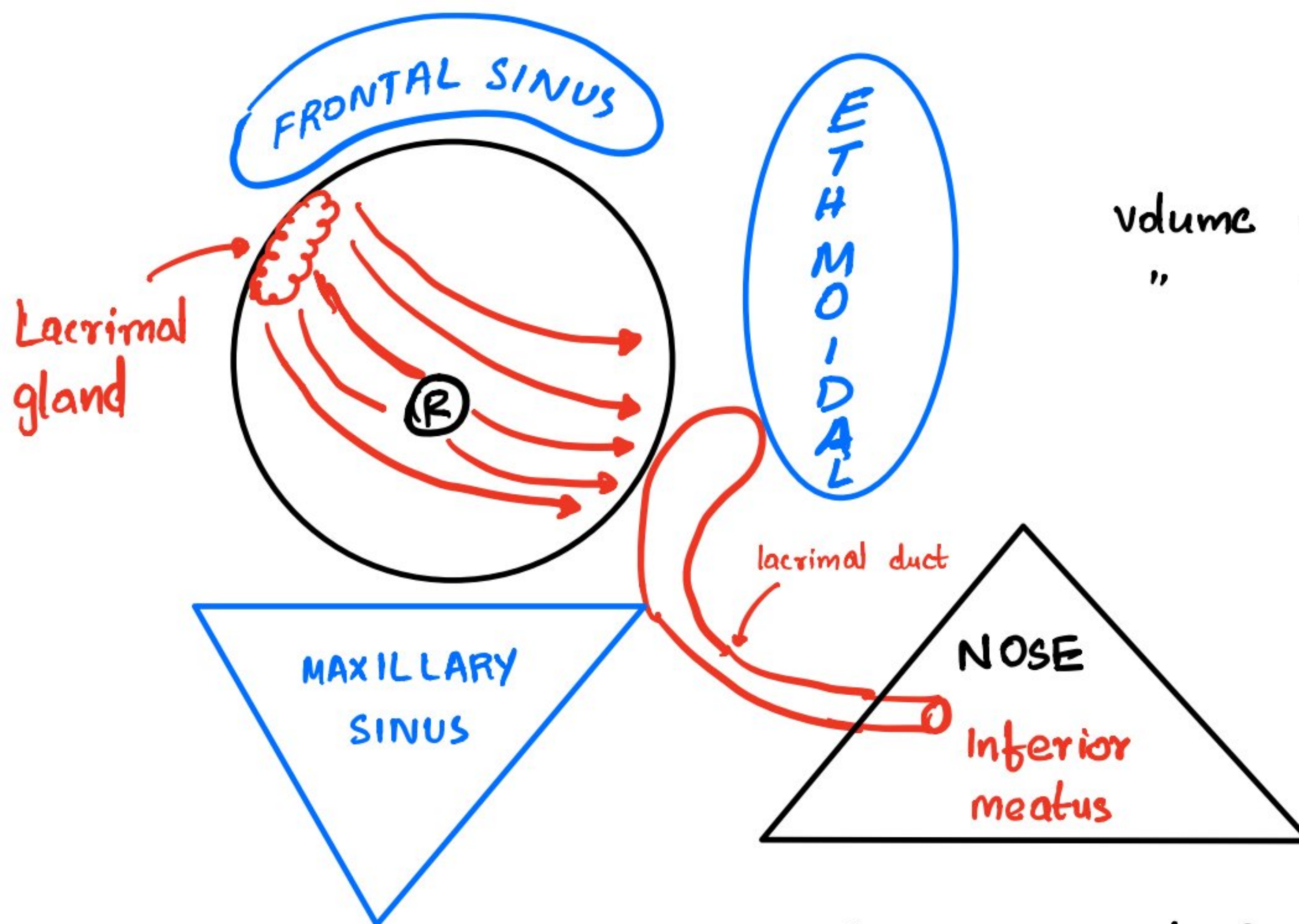
Branch of Maxillary division of Vth Nerve (V_2)



Sensation of cheek.

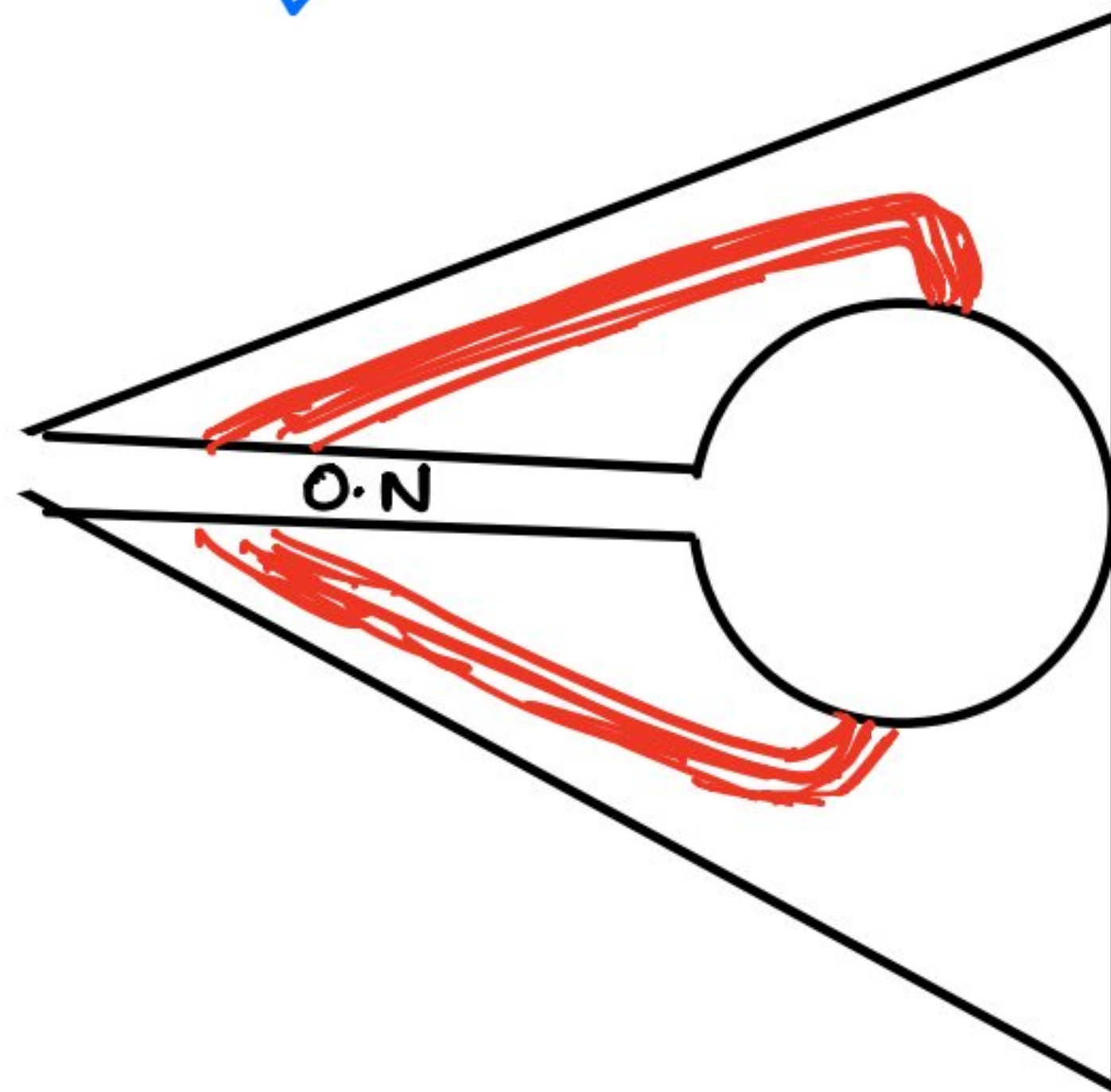


Relations of Orbit :



Volume of orbit = 30ml } Orbital fat
 " " Eyeball = 6.5ml }
 ↓

major component of orbit



Component of orbit :

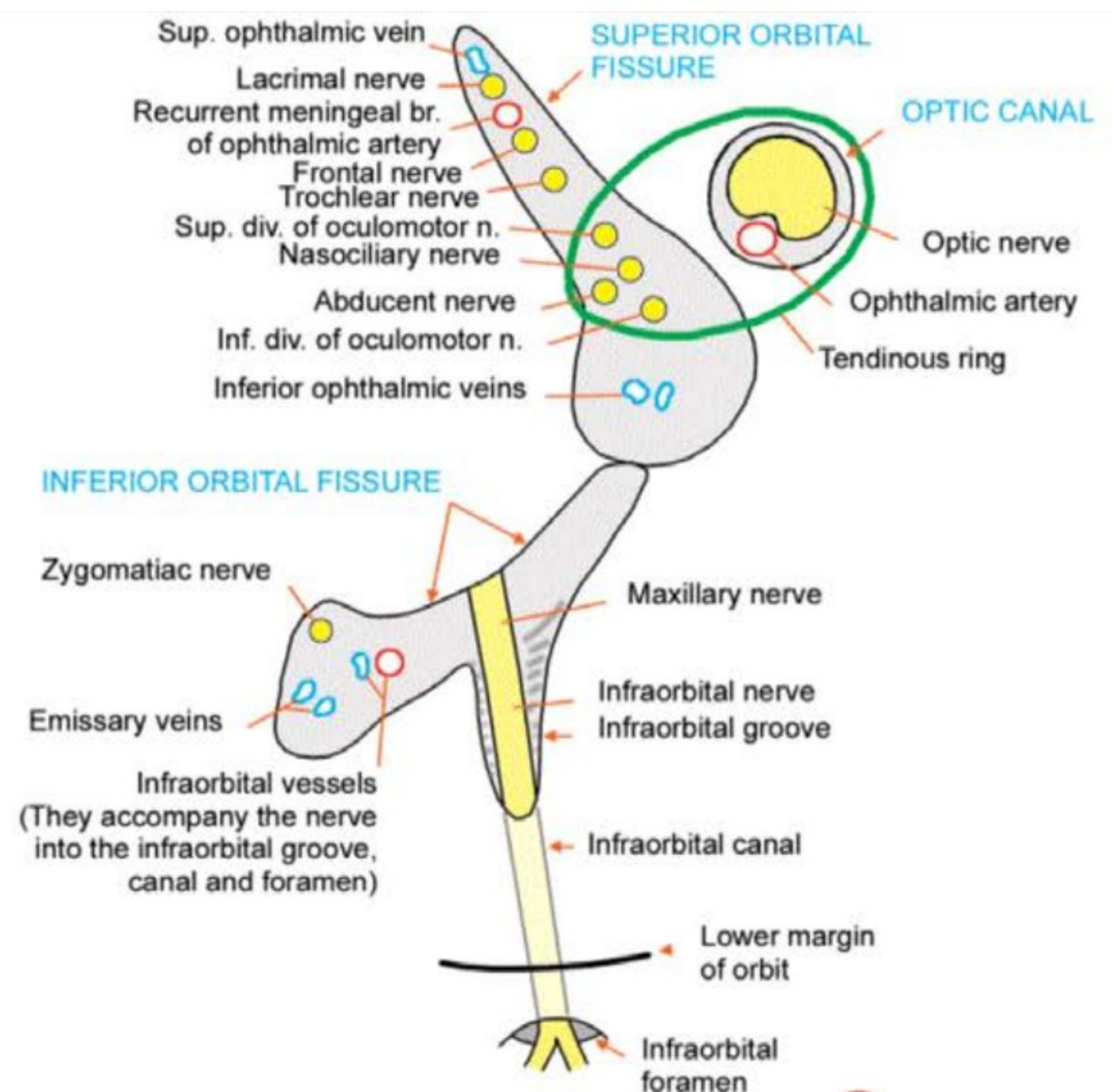
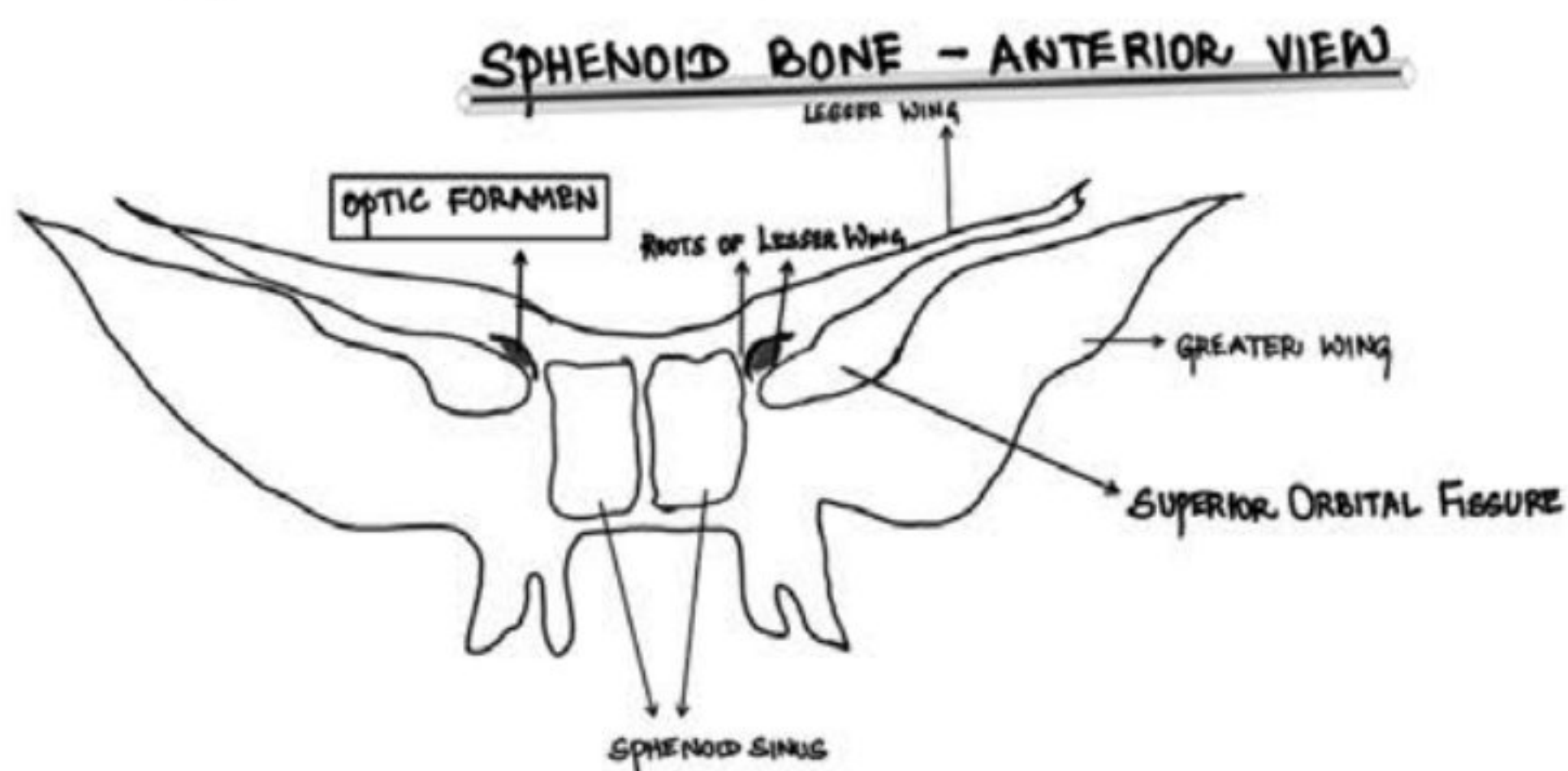
- 1) Eyeball
- 2) Optic Nerve
- 3) Orbital Fat
- 4) EOMs
- 5) Cranial Nerve
- 6) Vessels
- 7) Ciliary Ganglion
- 8) Lacrimal glands & sacs

Sphenoidal Bone

- Greater Wing
- Lesser Wing
- Body of Sphenoid

structure passing through optic foramen:

- i) Optic Nerve
- ii) Ophthalmic Artery



Sensory Supply of the eye:

Ophthalmic Nerve ← 1st Branch of Trigeminal N.

i) Lacrimal Nerve → **Lacrimal Gland**

- Conjunctiva
- Lateral eyelid

ii) Frontal Nerve →

- Conjunctiva
- Eyelid
- Part of Forehead

iii) Nasociliary

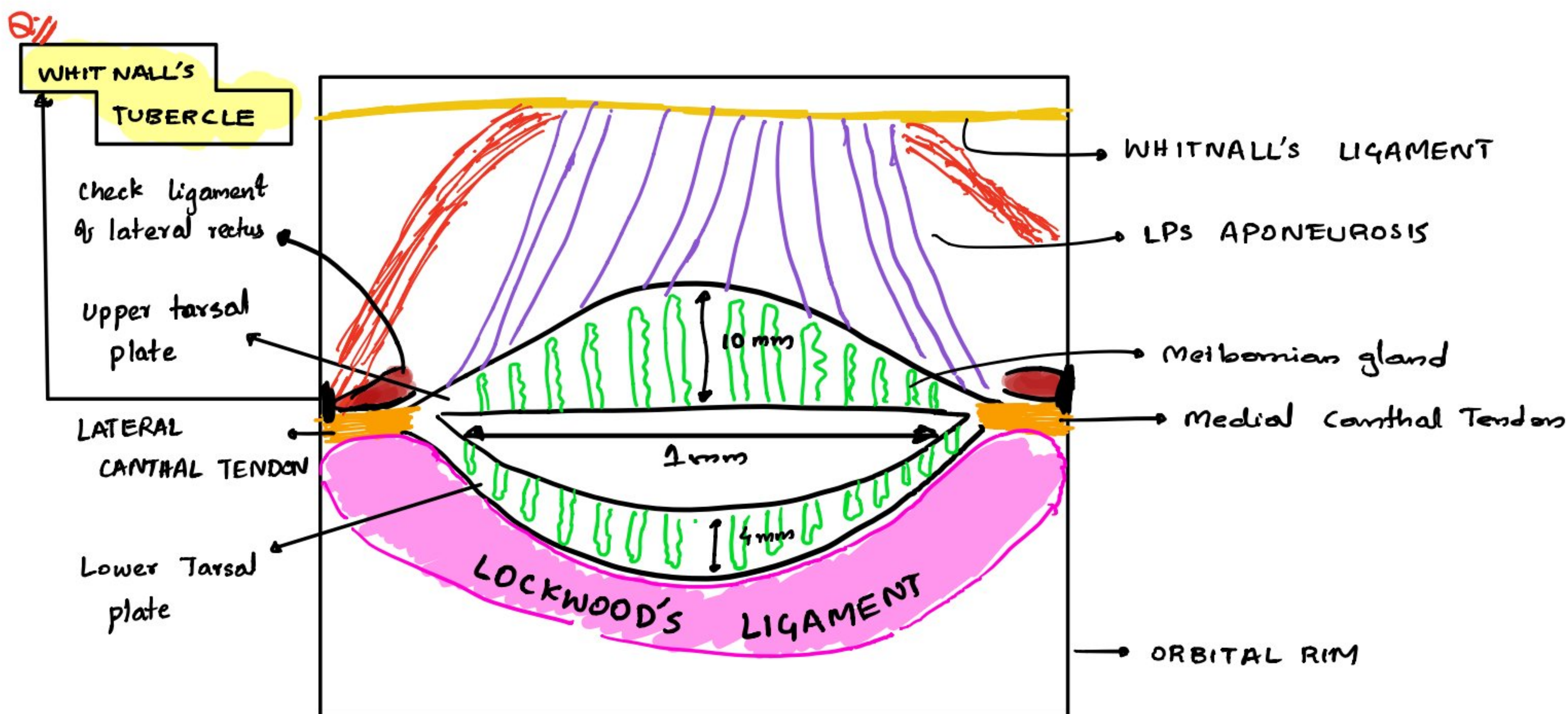
- long ciliary N. →
- Infra Trochlear N. →

- Cornea
- Ciliary body
- Iris

↓

- Conjunctiva
- **Lacrimal Sac**
- Caruncle
- Medial part of the Eyelids

Tarsal Plates:



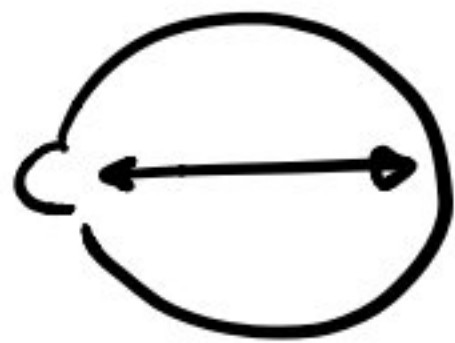
- Tarsal plates are firm, dense plates of **Connective tissue** & fibrous tissue?
- Check ligament of the lateral rectus is attached to **WHITNALL'S TUBERCLE**

Whitnall's Tubercle → small elevation on the orbital surface of the zygomatic bone, just within the lateral orbital margin.

structure attached to it:

- 1) Check ligament of lateral rectus
- 2) Lockwood's ligament {suspensory ligament of the eyeball}
- 3) Lateral palpebral ligament.

↳ Lateral horn of the Levator aponeurosis

	Eyeball of Newborn	Adult
Size (Axial length)	16.5mm (small eyelid → +2.5D Hypermetropia) $\approx$ Adult size attained @ 4yr → Emmetropization	$\approx$ 24mm 
Corneal diameter	10mm	12mm
Lens Shape	Spherical	Biconcave
Orbital Angle	50° → Divergent orbit	45°
Visual Acuity	< 6/60 • Fovea not completely formed {only after 3 months fully developed} # B&W upto 3 months • Refractive error of +2.5D $\approx$ 6/6 attained @ 6yrs	6/6

Children Milestone:

- 1) Fixation → see object continuously from one eye
 → 6wks = starts developing, completed by 3-6 months
 → unocular phenomenon.
- 2) Fusion → Binocular Phenomenon
 → see object using both eye
 → 6 months = developed, stereopsis starts developing
 (3D vision)
- 3) Stereopsis → Binocular phenomenon
 → 6 year = fully developed

Embryology of Eye:

• Surface Ectoderm:

S → Skin of eyelids / appendages

L → Lens

L → Lacrimal Apparatus

E → Epithelium of cornea / conjunctiva.

Neuroectoderm / optic Cup

R → Retina

Outer Lip = RPE (10th layer)

Subretinal Space

Inner Lip = Neurosensory → All 9 layers retina

O → Optic stalk / Optic Nerve

M → Muscles of Iris (Sphincter Pupillae
Dilator Pupillae)

E → Epithelium of Iris / Ciliary Body

Neural Crest:

Neural

- Optic nerve covering
- Myelin sheath
- Ciliary ganglion
- Melanocytes

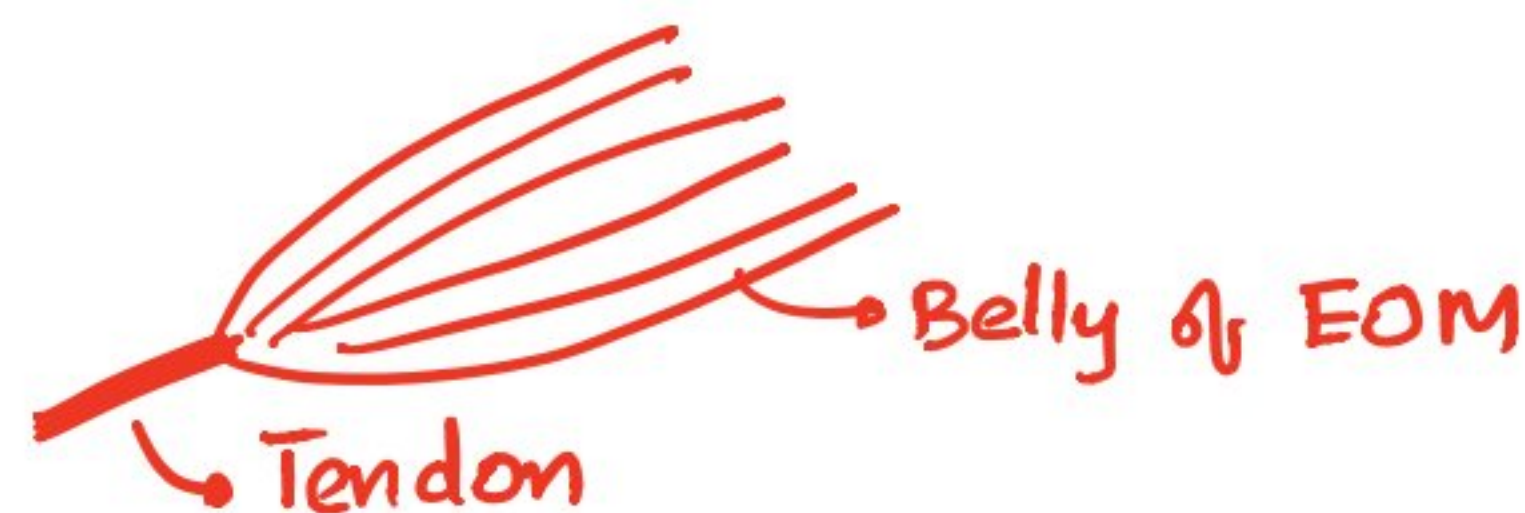
Q: Myelin sheath of O.N formed by Oligodendrocytes

Mesoderm

- 1, Choroid
- 2, Hyaloid Artery (1^o vitreous)
- 3, Belly of E.O.M
- 4, Temporal portion of sclera

Mesenchymal

- Cornea except epithelium
- Sclera except epithelium
- Orbital bones
- Trabecular meshwork
- Schlemm's Canal
- Ciliary Body muscle
- Iris / Ciliary Body stroma
- Tendons of EOM

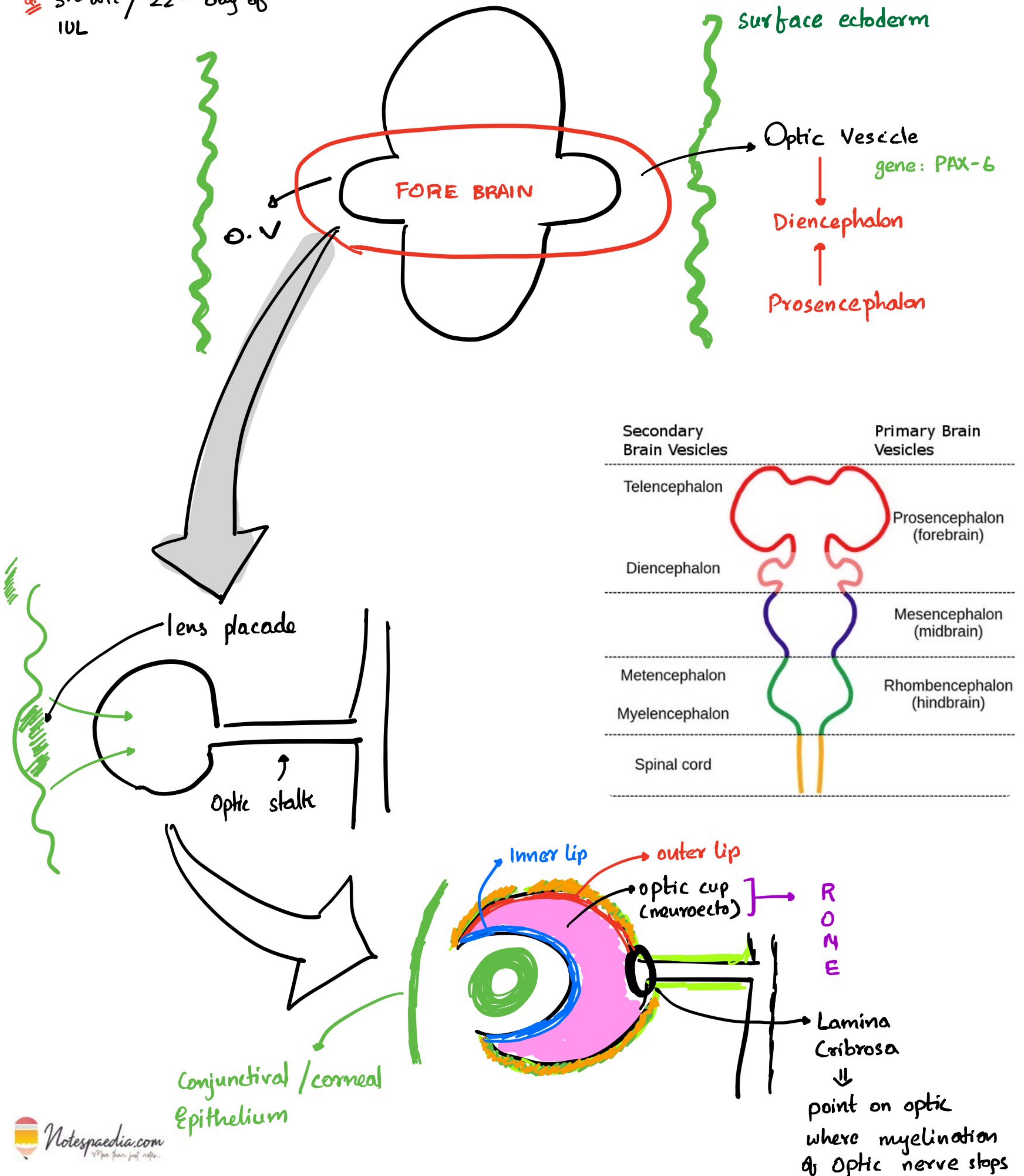


Period after conception	Major milestone
Q: 3 wk	→ • Optic groove appears
4th wk	→ • Optic pit develops into optic vesicle • Lens plate forms • Embryonic fissure develops
1 st month	→ • Lens pit develops, followed by formation of the lens vesicle • Hyaloid vessels develop
Q: 4th month	→ • Formation of retinal vasculature begins • Hyaloid vessels regress Q: • Canal Schlemm • Formation of physiological optic disc cup and Lamina Cribrosa

5th month

- Bowman's membrane develop
- Formation of the major arterial circle and sphincter muscle of the iris
- Photoreceptors differentiate
- Eyelid separation begins

3rd wk / 22nd day of IUL



Peter's Anomaly:

- where lens vesicle don't detach properly from the surface ectoderm
- lens have attachment to the corneal / conjunctival epithelium → corneal opacity



- D/D for Hazy cornea @ birth.
- (MC) congenital cause for penetrating keratoplasty in children



Full thickness corneal transplant

Lacrimal gland (main)

SURFACE EPITHELIUM

LID GLANDS



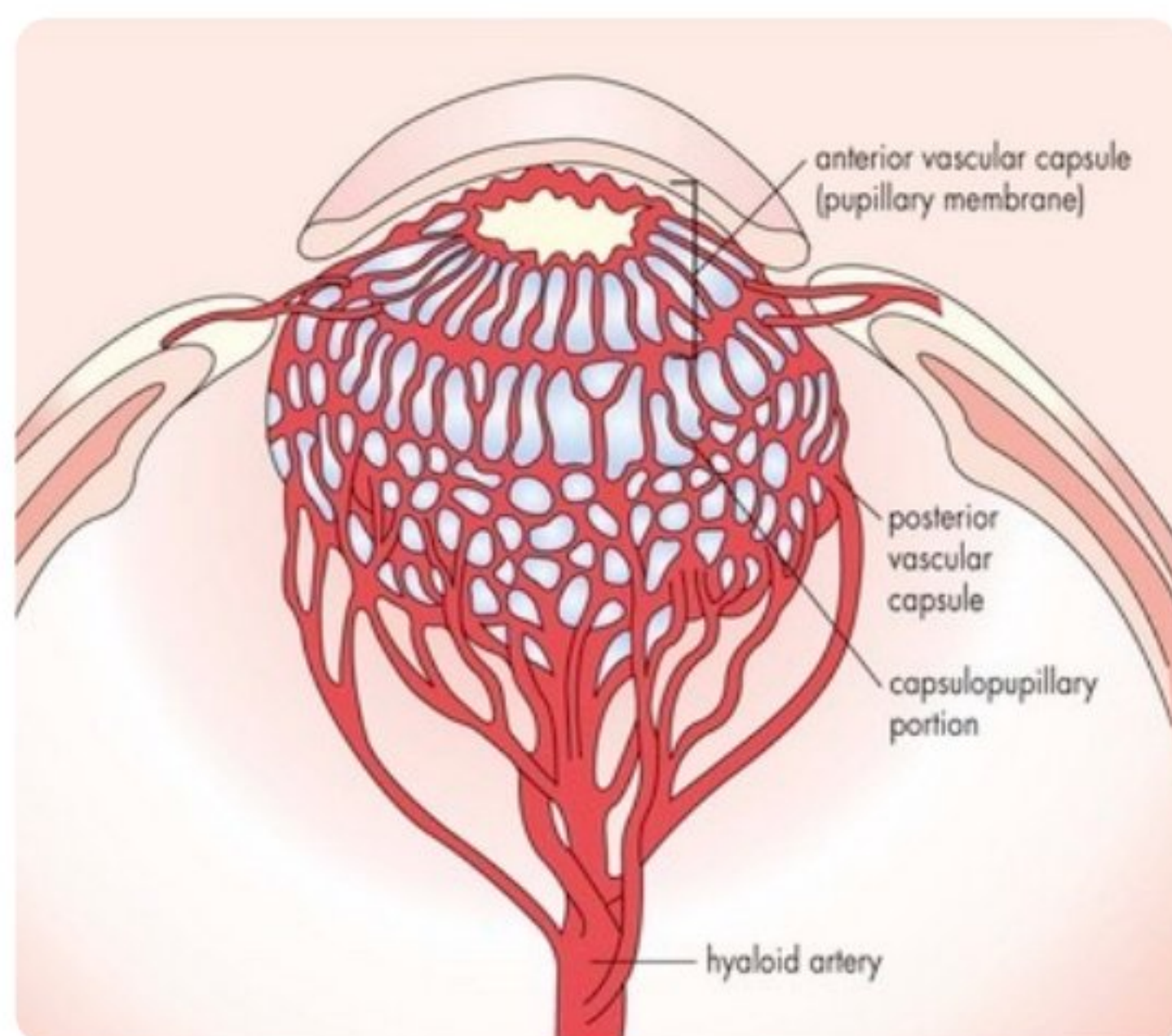
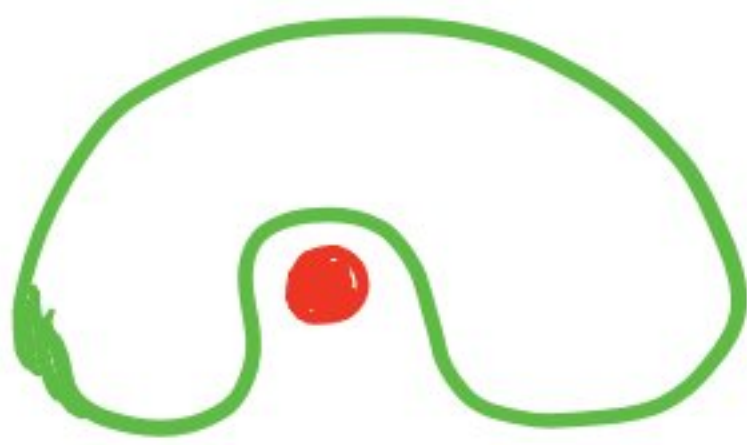
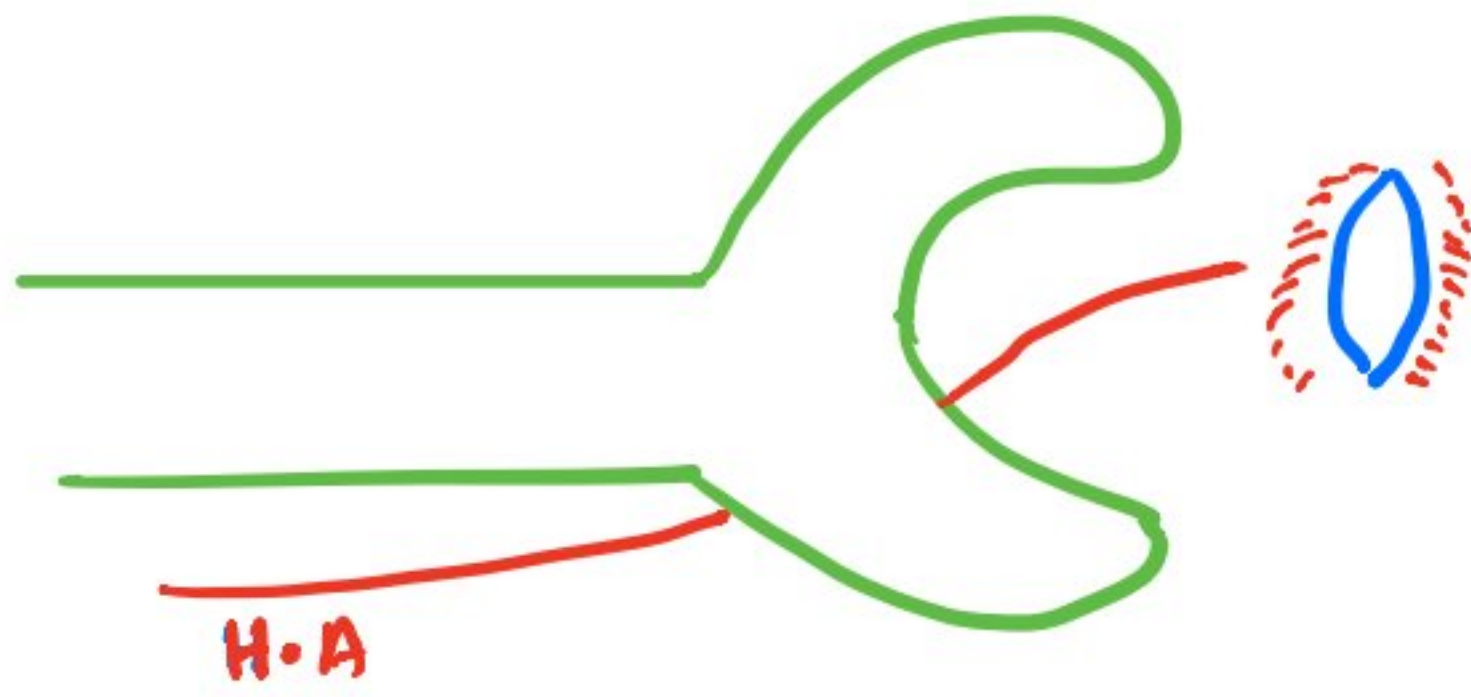
- ① GLANDS OF MOLL
- ② MEIBOMIAN GLAND
- ③ GLAND OF ZEIS

LACRIMAL GLANDS



- ① KRAUSE
- ② Goblet cells
- ③ HENLE
- ④ WOLFRING
- ⑤ MANZ

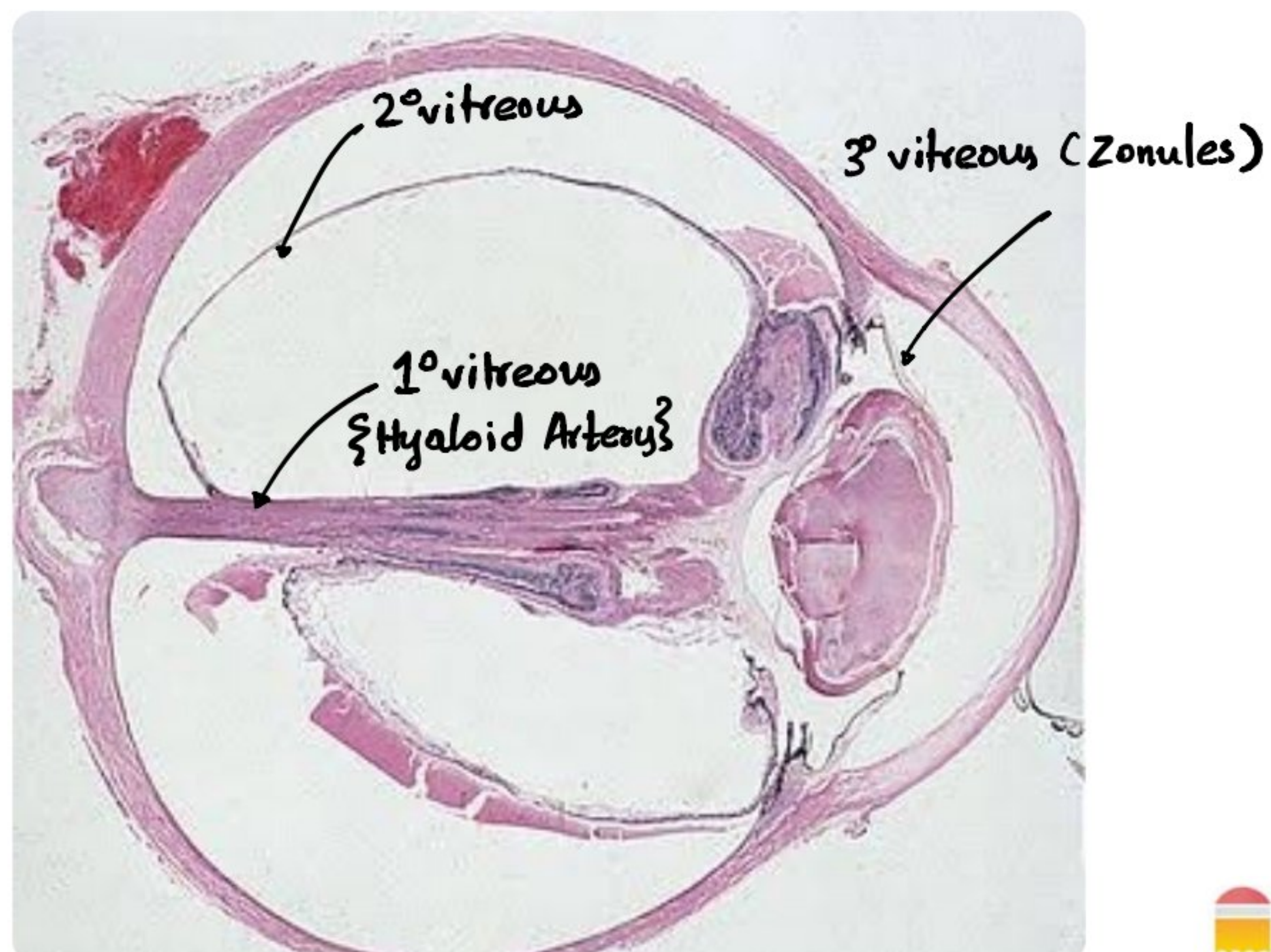
Hyaloid Artery.



Tunica vasculosa lentis

Remnants of Hyaloid Artery

- 1) Muscae Volitans
- 2) Mittendorf's dot
- 3) Bergmeister Papilla



PHPV

Persistent Hyperplastic 1° vitreous (PHPV)

- Poor prognosis

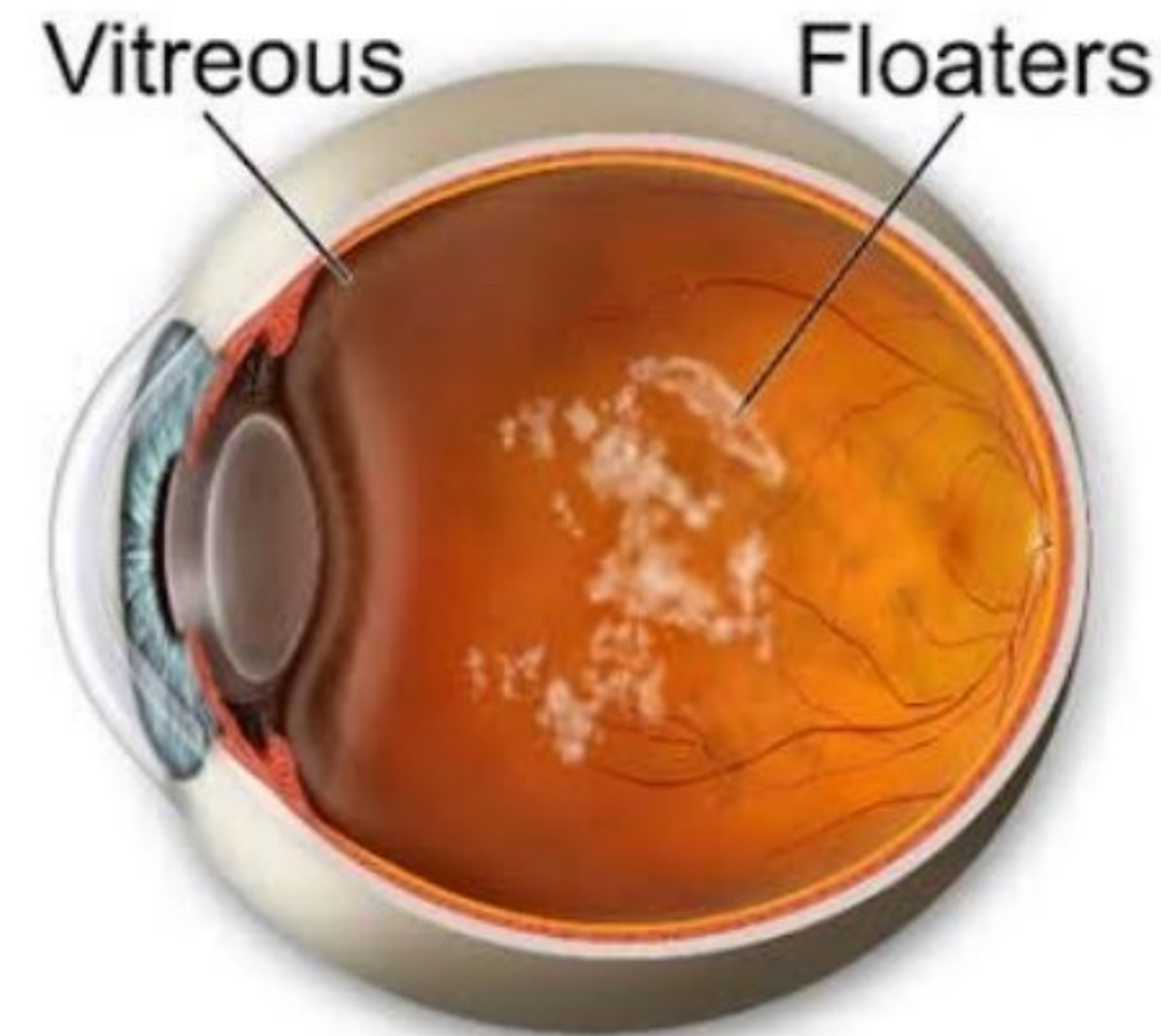
q/w i) U/L

- ii, micro - opthalmus
- iii, Cataract (attach to lens)
- iv, Retinal detachment (attach to lens)
- v, Vitreous Hmage

• **PATAU SYNDROME (Trisomy 13)**

Muscae Volitantes {clouds}

- Debris of Hyaloid vessels in vitreous
- Multiple floaters
- Positive scotoma



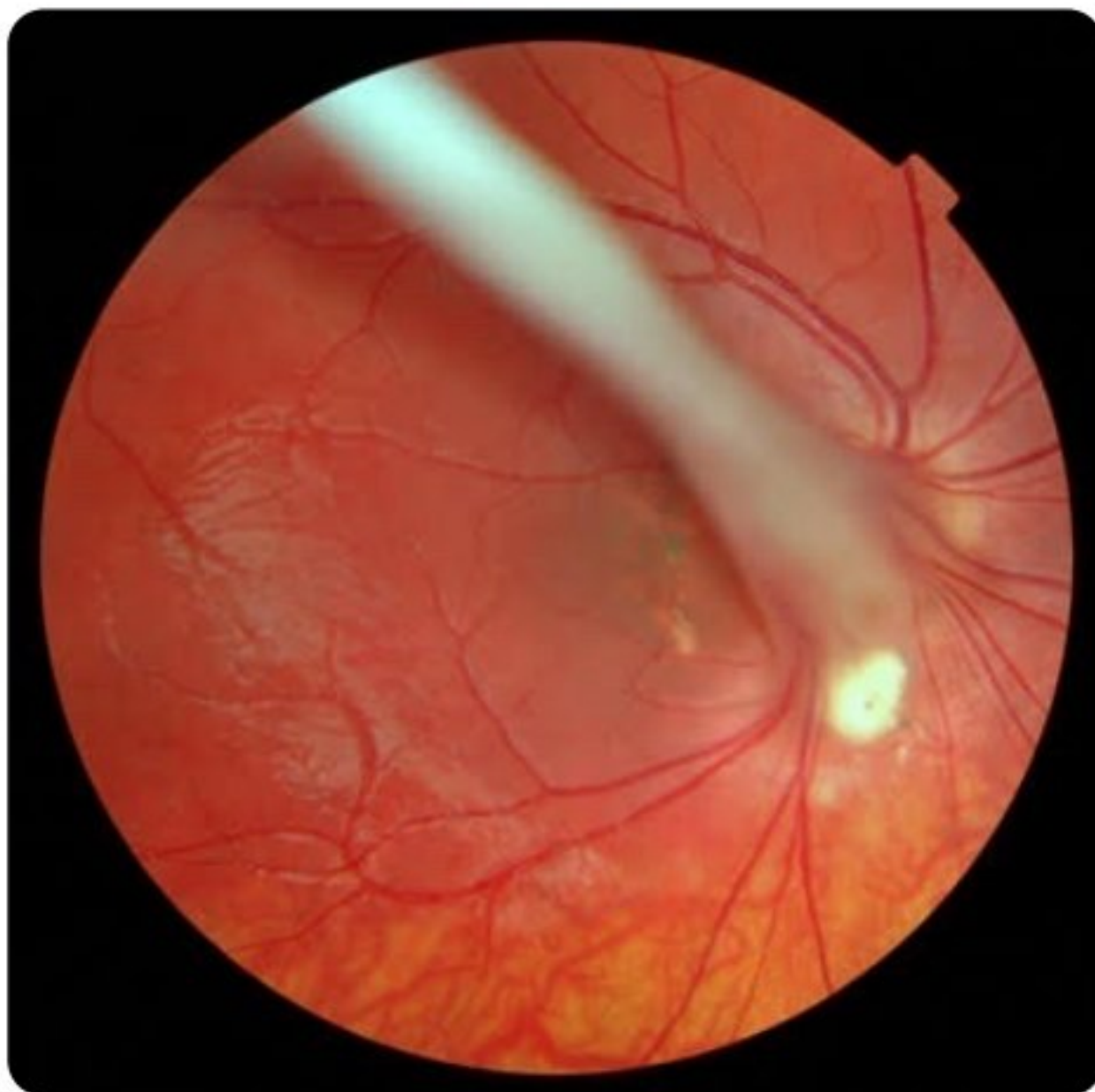
Muscae volitantes

- Mitten Dorf dot → Ant. segment of Hyaloid vessels attached to lens.
→ asymptomatic
→ Good prognosis



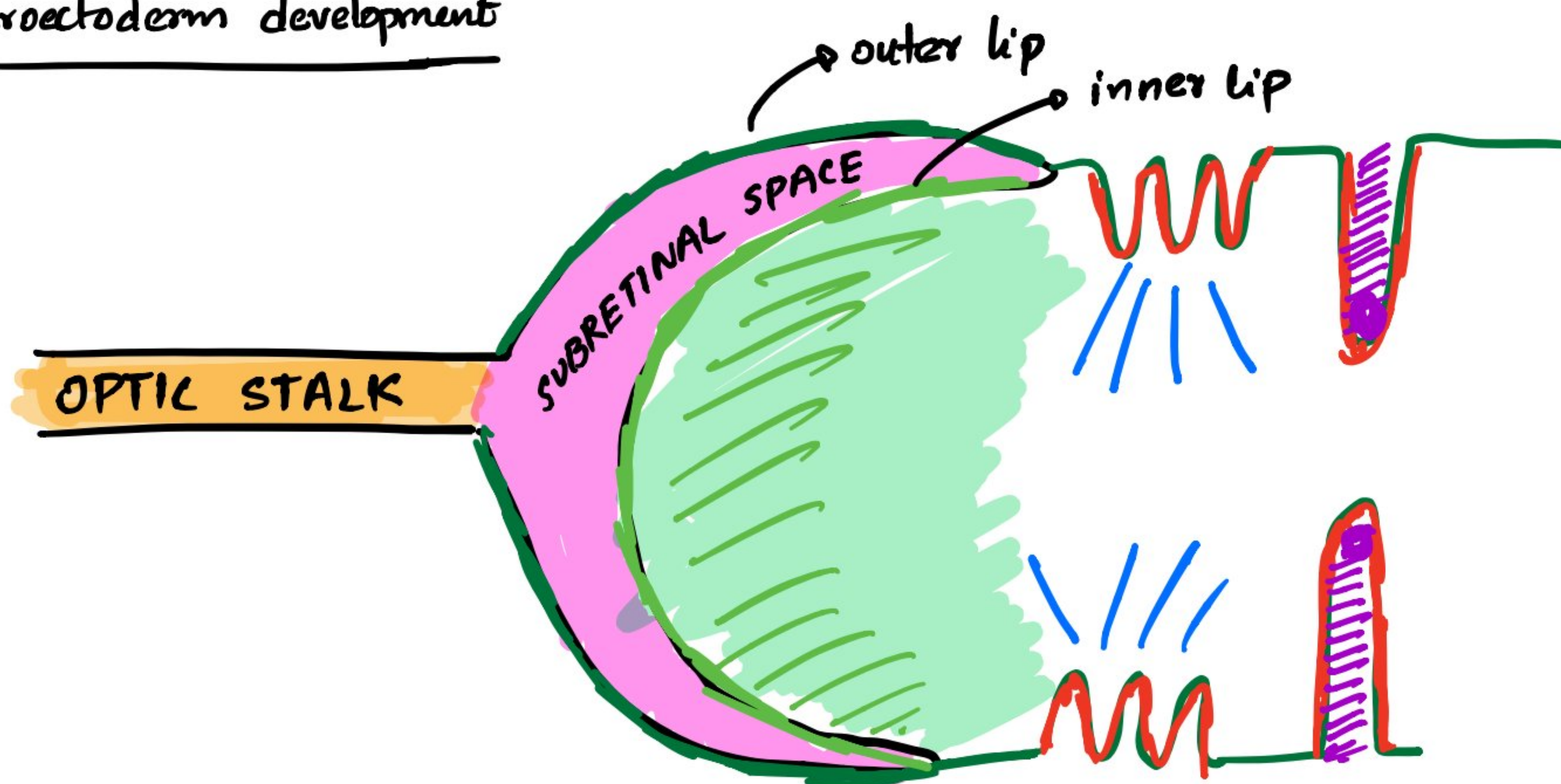
Mitten Dorf dot

- Bergmister Papilla → Remnant of Hyaloid vessels post segment
→ Asymptomatic
→ Good Prognosis



Bergmister Papilla

Neuroectoderm development



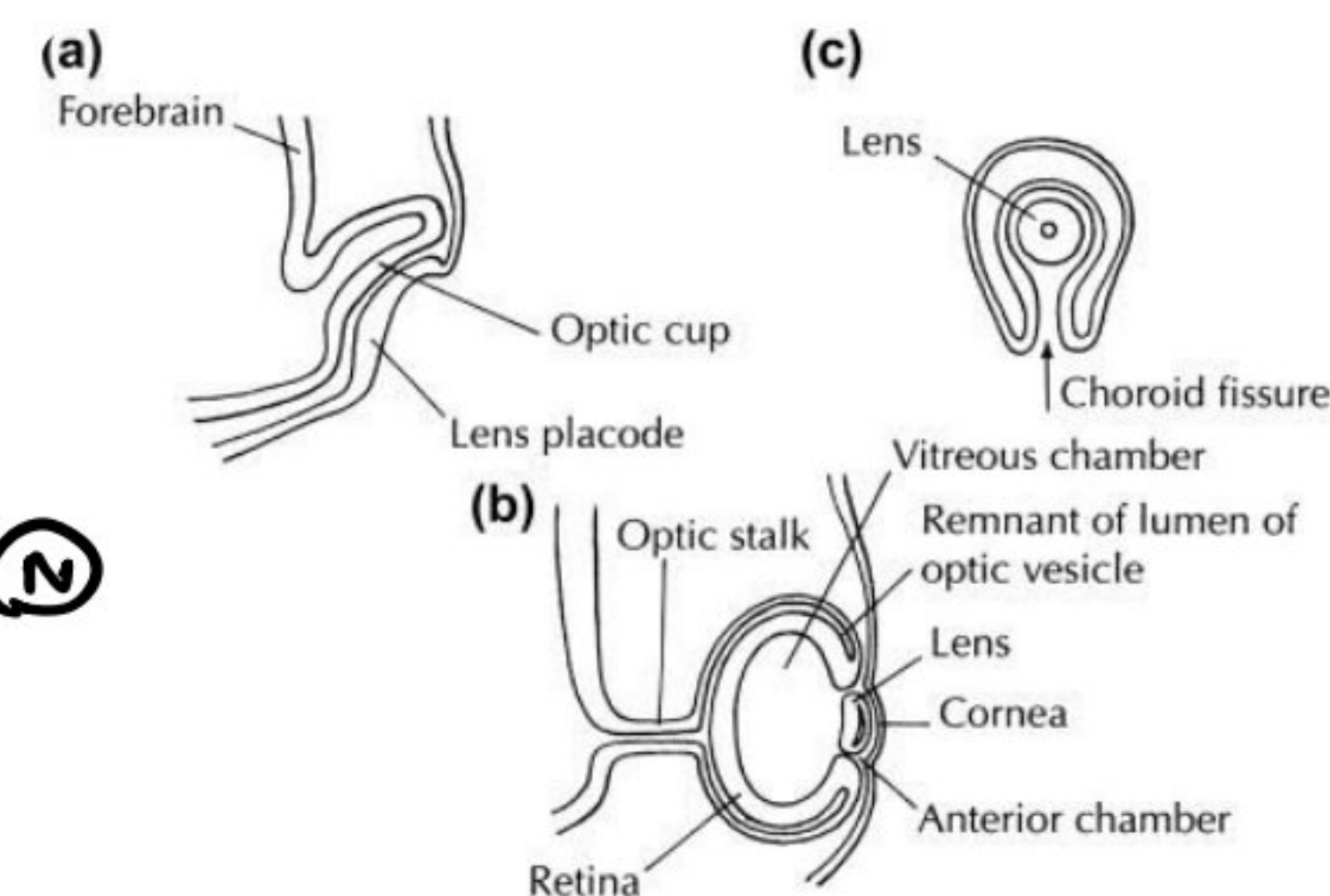
- 1) Retina → a) RPE from outer lip of optic cup
b) Neurosensory retina from inner lip of optic cup of all 9 layers.

- 2) Optic Nerve
- 3) Ciliary Epithelium
- 4) Iris Epithelium
- 5) Sphincter Epithelium
- 6) Dilar papillae
- 7) Zonules (3° Vitreous)
- 8) Vitreous (2° Vitreous)

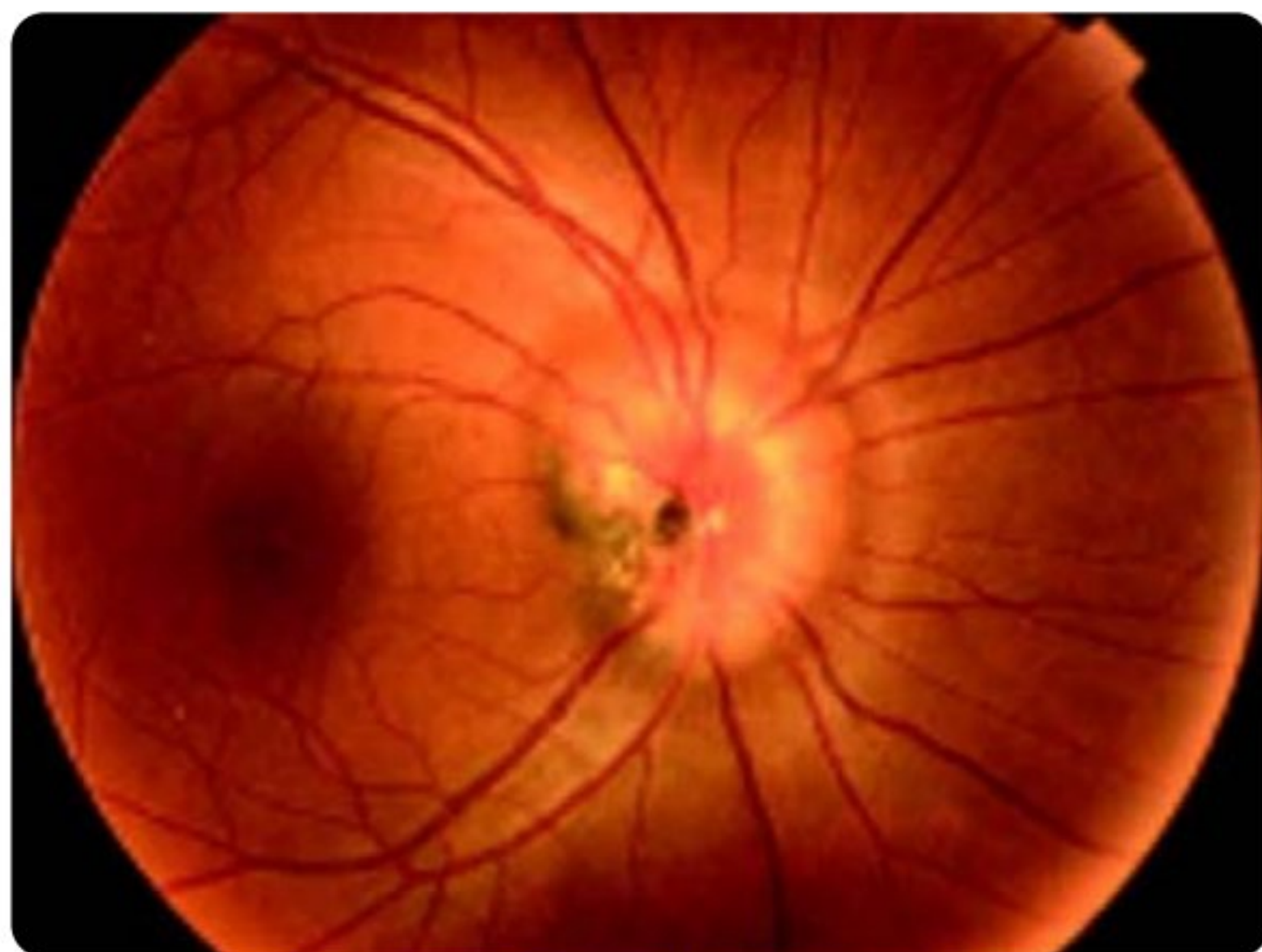
COLOBOMA:

• Embryonic fissure fuses @ 36 days / 5wk of Gestation → (N)

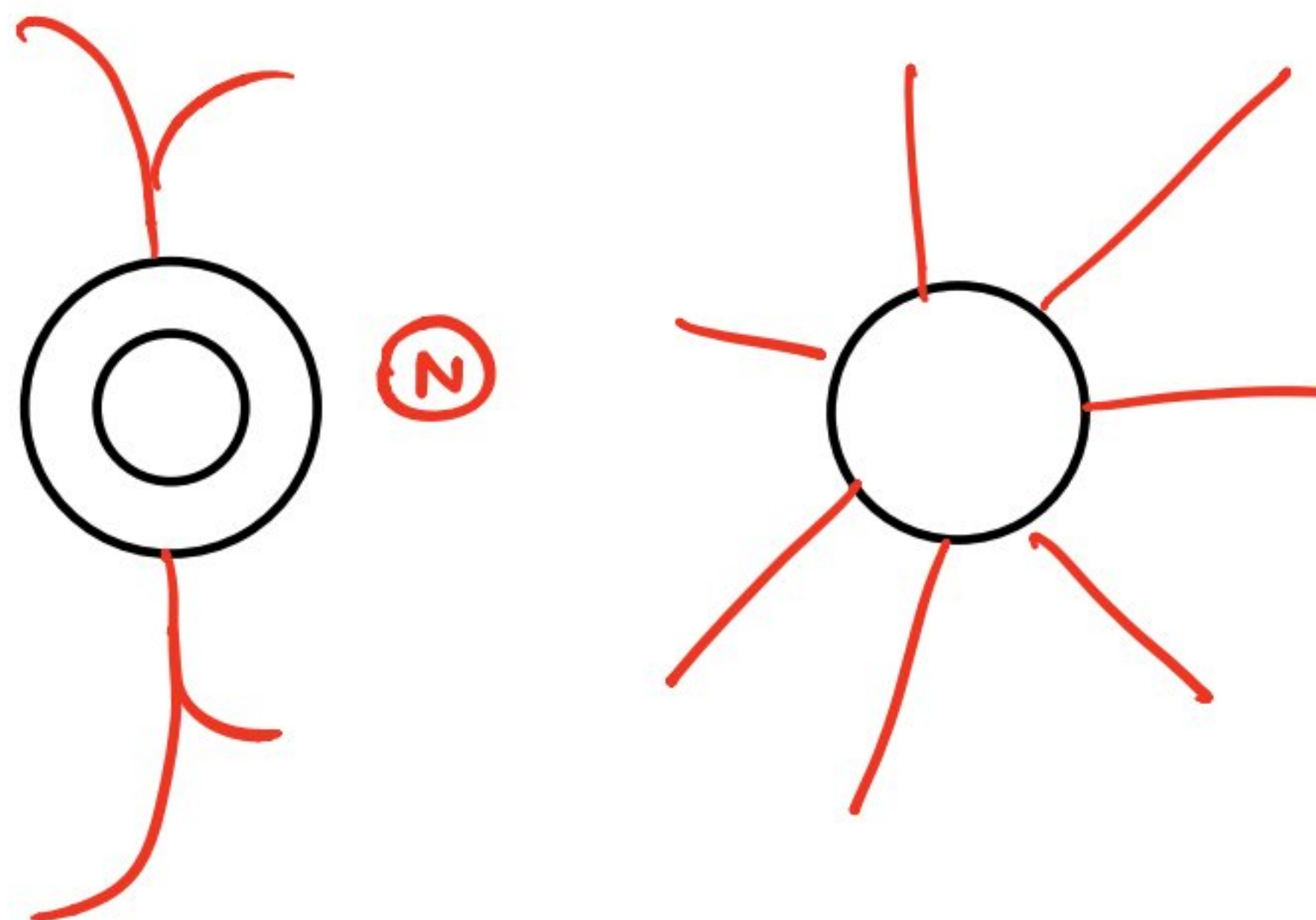
• Failure of closure → **COLOBOMA**
(MC) location: Inferonasal



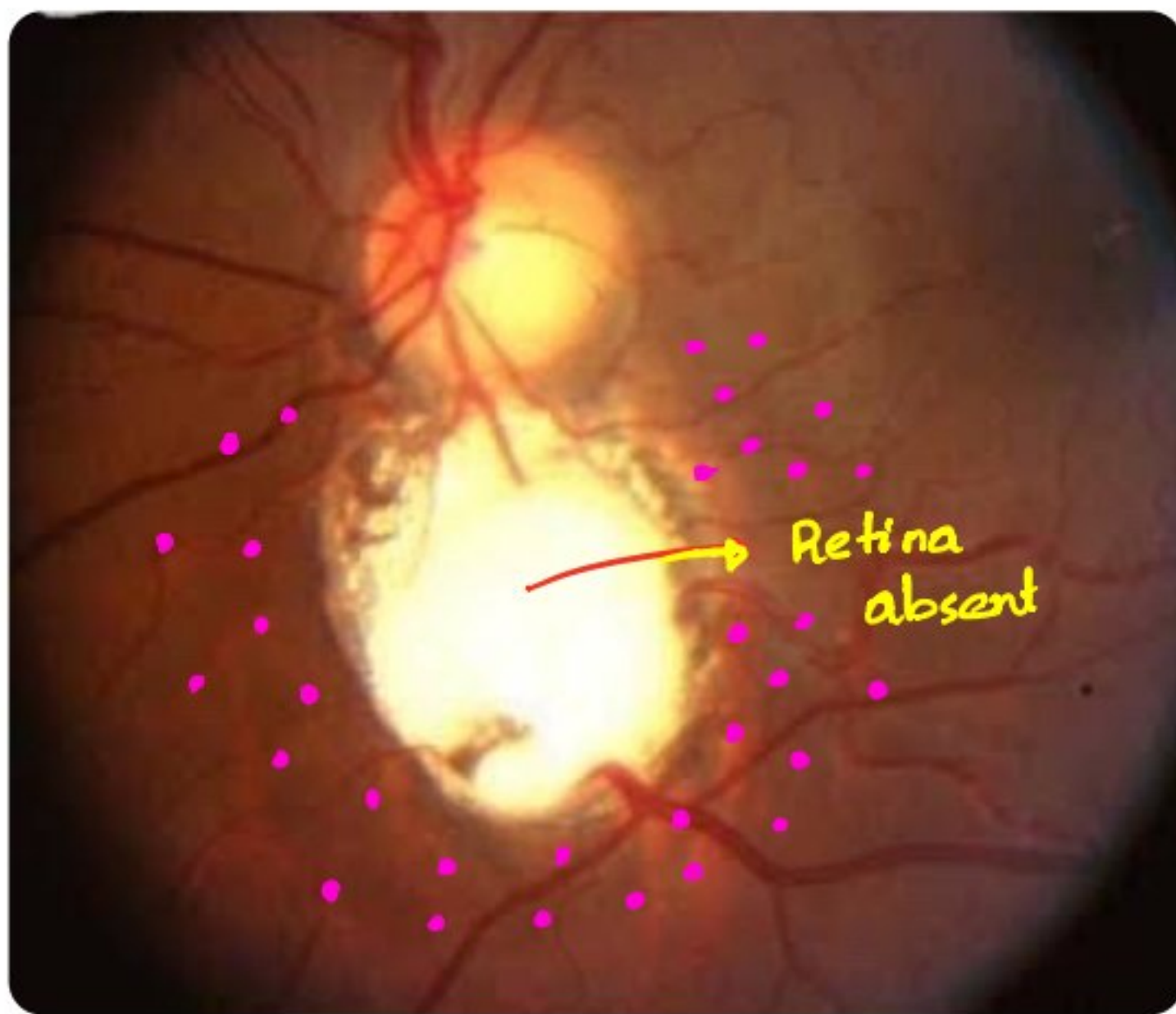
Coloboma of Optic disc



Morning glory disc



ii) Chorioretinal Coloboma:



- can lead to Retinal detachment
- Rx: **BARAGE LAZER** with 3 layers of spot around the defect to prevent detachment.

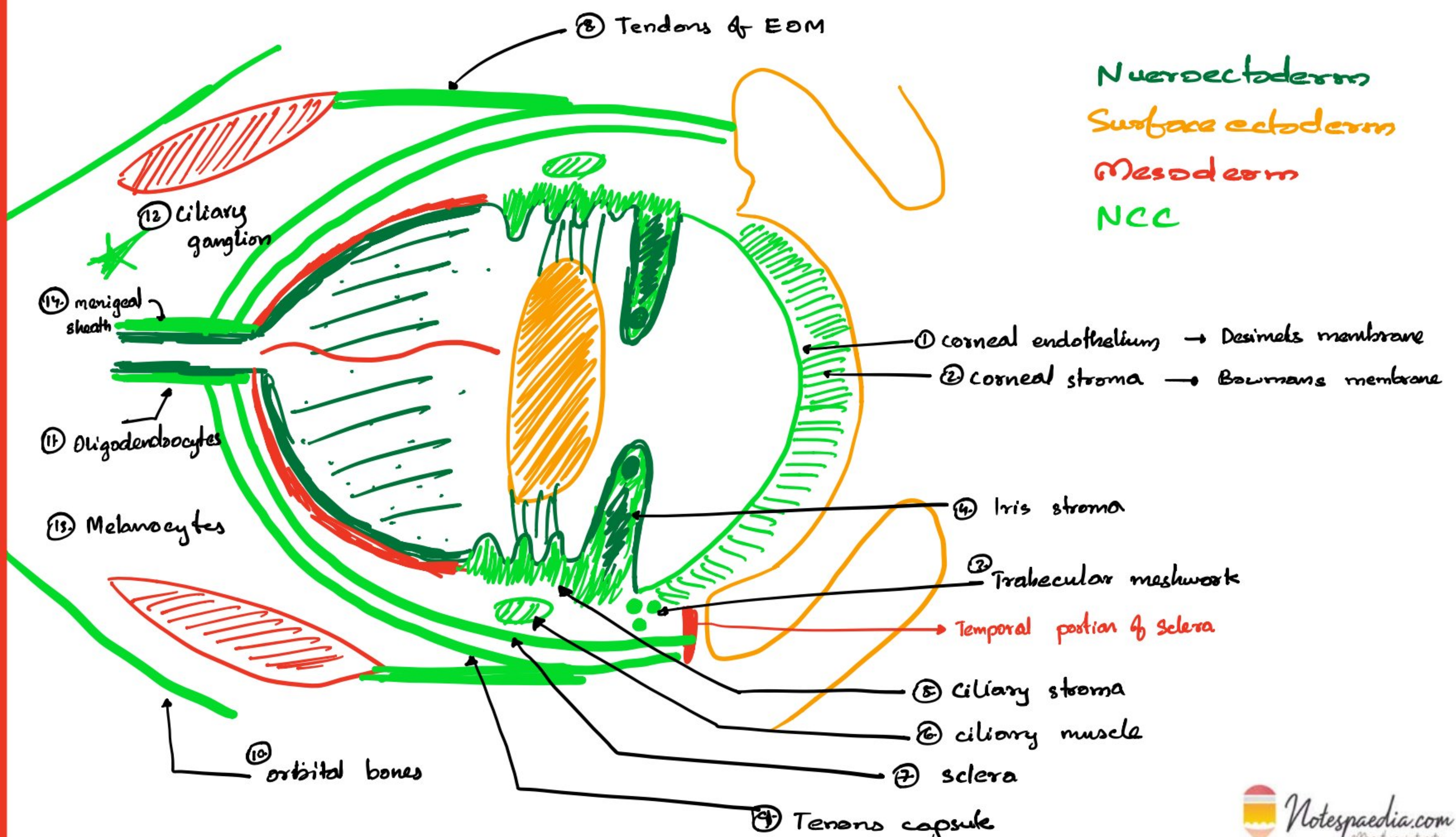
iii) Iris Coloboma:

- Pear shaped pupil / Tear shaped
- a/w lens subluxation → d/t weak zonules



- (MC) location is **inferonasal quadrant**

NEURAL CREST CELLS DERIVATIVES



Sclera $\Rightarrow$ **NCC** + Mesoderm

EOMs $\Rightarrow$ **Mesoderm** + NCC

Arrest / failure of NCC migration:

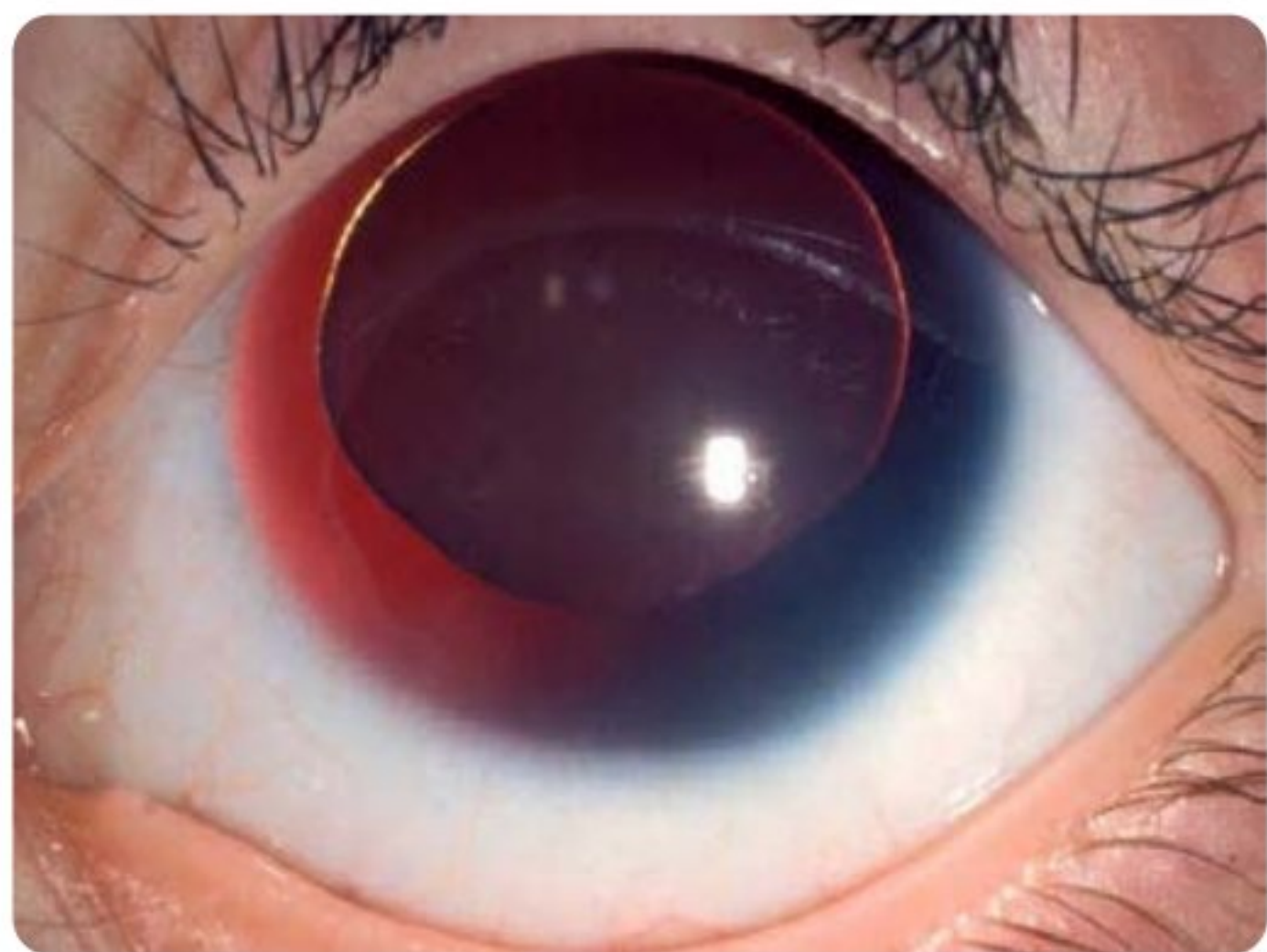
Congenital Horner's Syndrome: Sympathetic palsy & sympathetic stimuli required for migration of NCC migration?



- 1) Miosis
 - 2) Mild ptosis
 - 3) Enophthalmos
 - 4) Hypopigmented Iris $\rightarrow$ Pathognomonic for congenital Horner's
- { HETEROCHROMIA }

Neurocristopathies: Inappropriate interaction b/w NCC & Neuroectoderm

Congenital Aniridia:

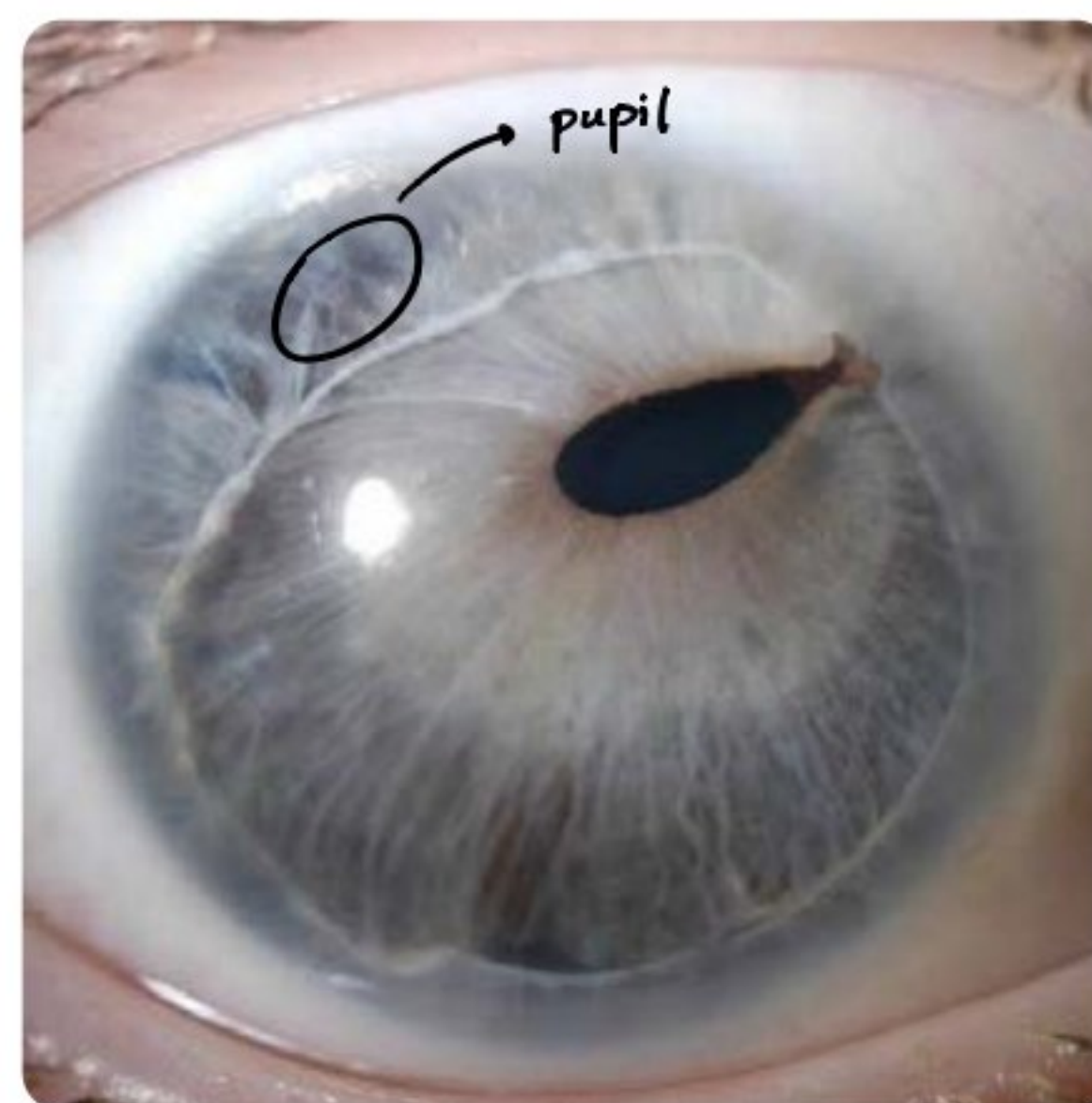


• d/t to mutation in **PAX 6 gene**

$\downarrow$
required for proper interaction
b/w NCC and neuroectoderm

Axzenfeld Rieger $\rightarrow$ "AD"

- Anterior segment dysgenesis
- Eccentric Pupil & corectopia
- Polycoria & Multiple Pupil
- TM not well developed $\rightarrow$ developmental glaucoma
- Iris stroma dystrophy.



BASIC INVESTIGATION:

1) Visual Acuity:

SNELLEN'S CHART

- pt @ 6m
- Subjective test

Minimal Angle of Resolution (MAR)

$\hookrightarrow$ smaller the angle better is the vision
minimal angle { 6/6 } = 1 minutes of arc

MAR = 1 / visual acuity

E

$\rightarrow$ Pt. seen only @ 6m

6/60

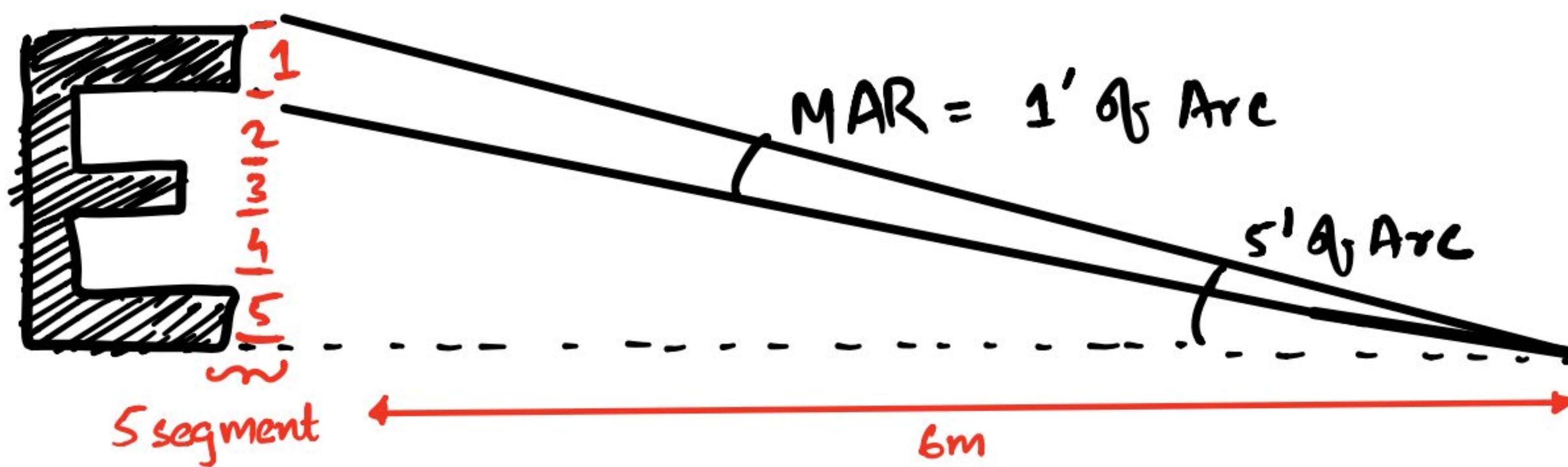
$\rightarrow$ (N) Individual can see this @ 60m

6/36 $\Rightarrow$ MAR = 6' of arc

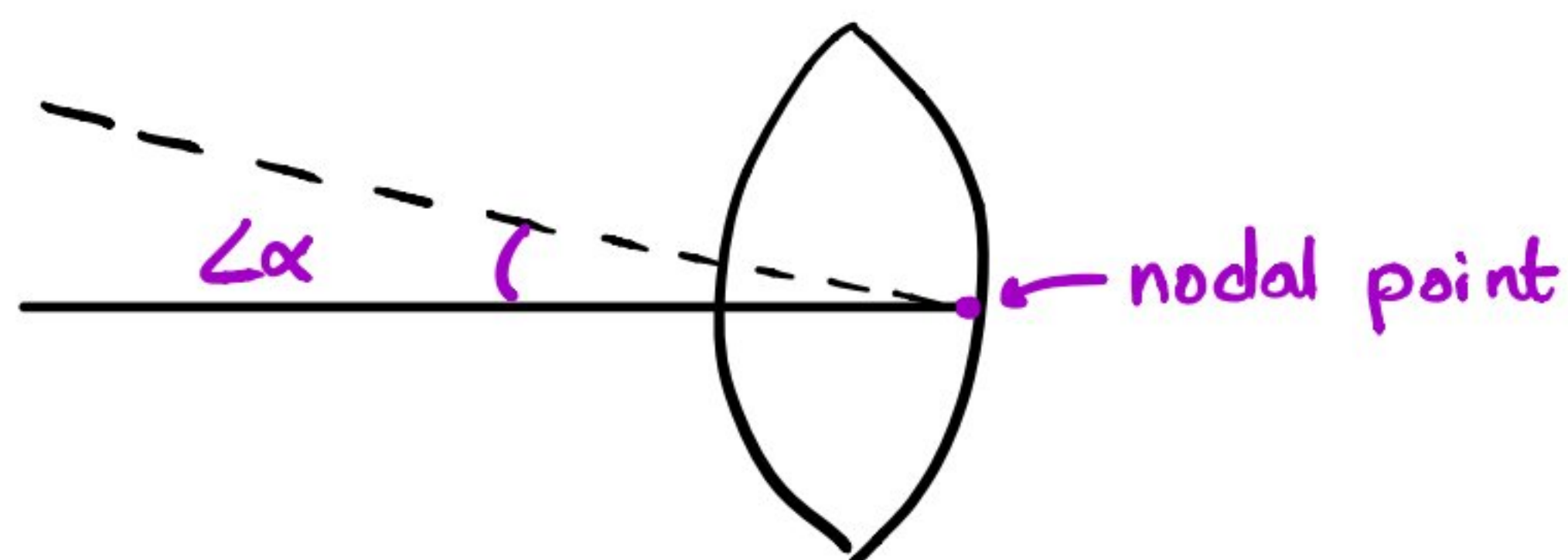
6/24 $\Rightarrow$ MAR = 4' of arc

$\vdots$
 $\downarrow$
etc.

Total angle $\rightarrow$ MAR $\times 5$



Angles subtended on post. surface of lens $\Rightarrow$ NODAL POINT
 $\Rightarrow \angle \alpha$

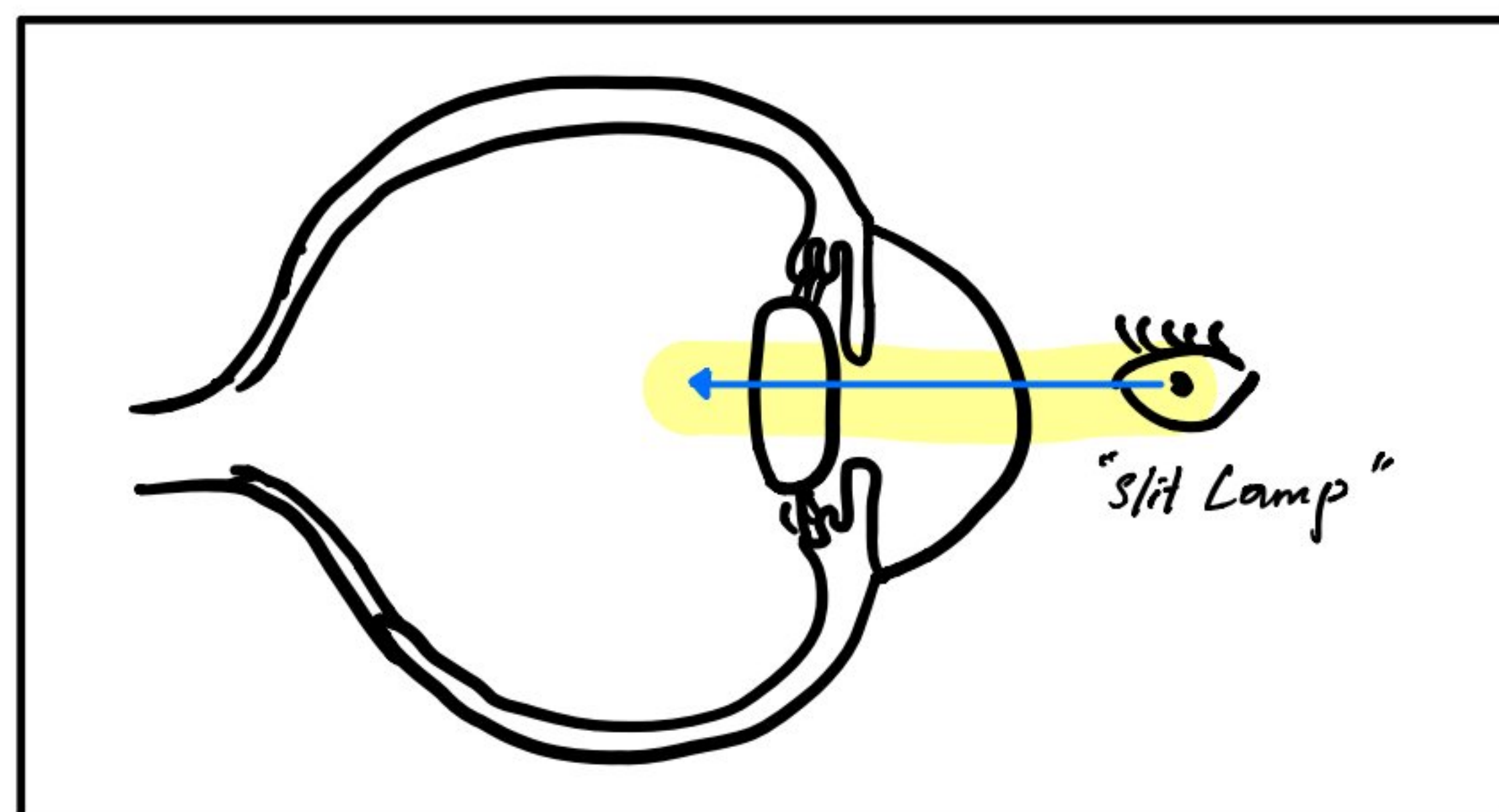
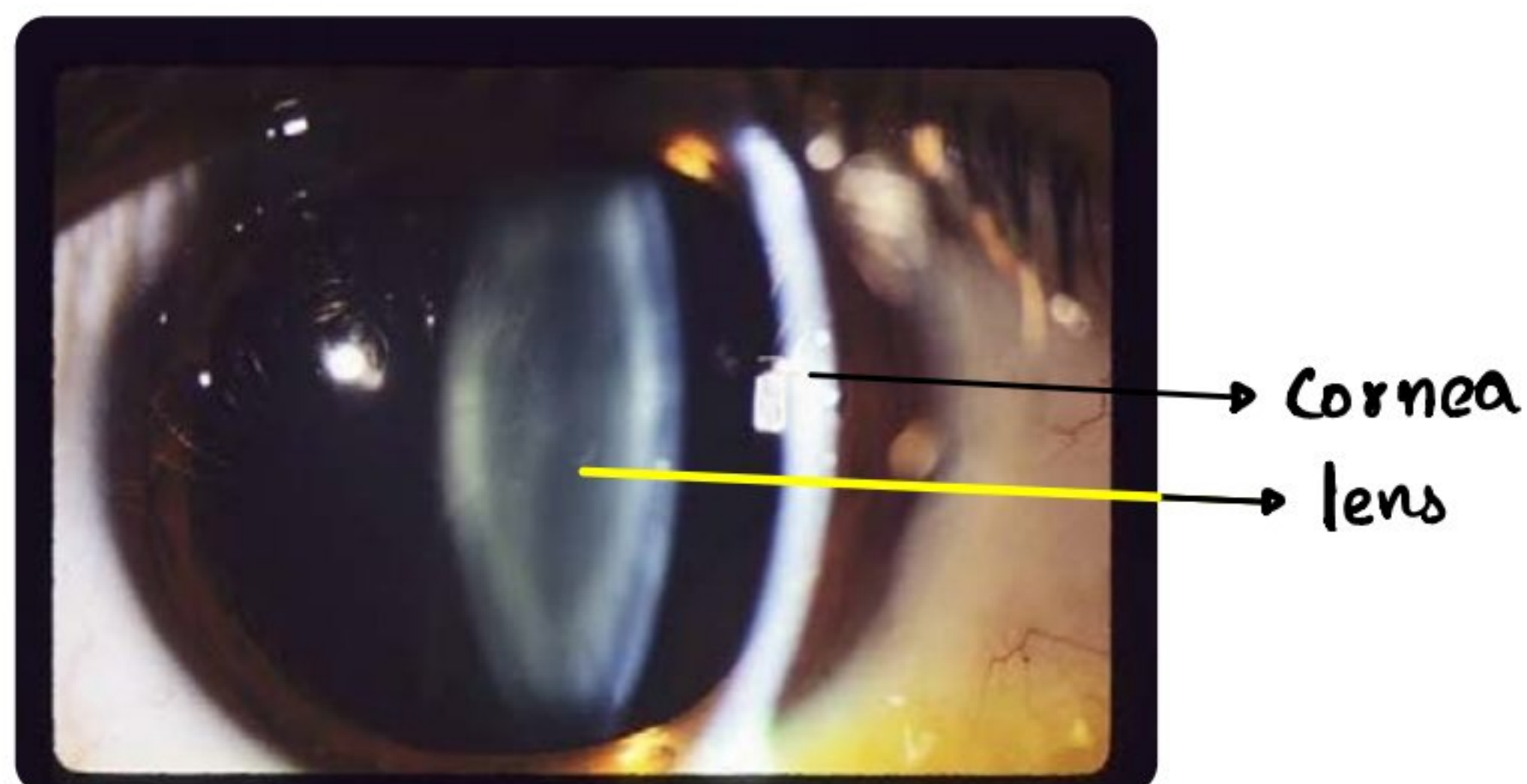


$6/60^x \rightarrow 5/60^x \rightarrow 4/60^x \rightarrow 3/60^x \rightarrow 2/60^x$

Projection of rays (which quadrant) $\leftarrow$ Perception of light $\leftarrow$ Hand movement $\leftarrow$ Counting fingers

2, Slit Lamp Examination

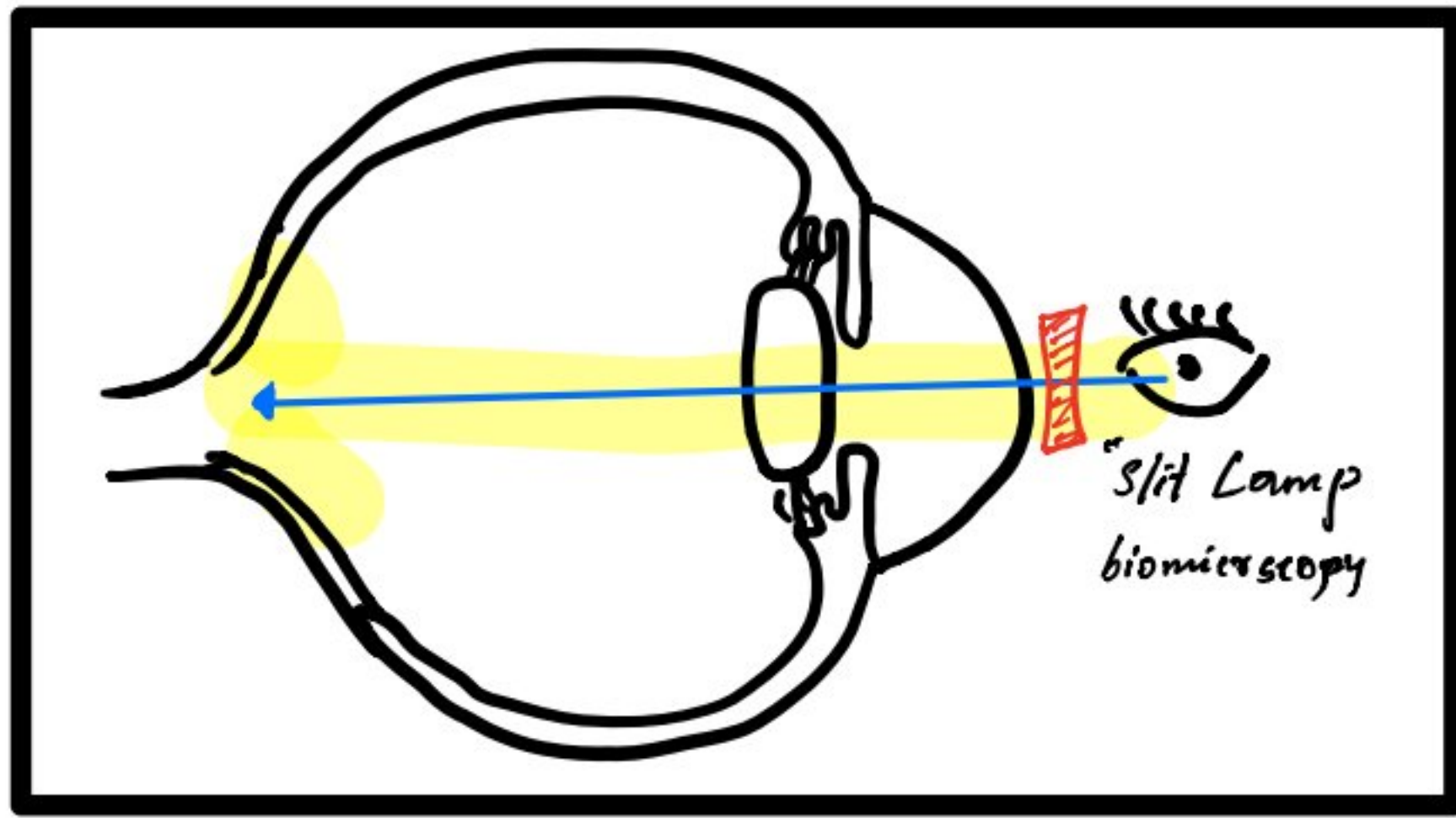
- See up to Ant. $1/3^{\text{rd}}$ of vitreous
- Binocular test & use both eyes $\rightarrow$ stereopsis 3D vision of eye
- Advantage: depth can be assessed



3, Slit lamp Examination $\Rightarrow$ Slit lamp + lens

Gold standard to see $\rightarrow$ Optic disc
 $\rightarrow$ Macula & central Retinae?

Lenses used in the investigation ——— $\oplus$ - 90D : see as Real inverted images



$\ominus$ - 58D
Hruby lens
Virtual erect image

4) Ophthalmoscope:

a) Direct Ophthalmoscope (unioocular)

i) Distant direct ophthalmoscope

- Doctor stands @ 25cm
- $\textcircled{N}$ Reflex $\Rightarrow$ RED ξ from vascular tissue choroid



- White reflex ($\textcircled{A}$) $\rightarrow$ Leukocoria

uses: i) Identify media opacity
ii) Refractive error assessment [BRUCKNER'S TEST]

iii) Patency of peripheral iridotomy.



Leukocoria

ii) Direct Ophthalmoscopy $\rightarrow$ Doctor comes as close to the patient.

- visualise $\rightarrow$ Optic disc ξ Macula
- Unioocular test $\rightarrow$ No depth visualised ξ no stereopsis
- Virtual erect image

- $\textcircled{R}$ eye of doctor $\rightarrow$ $\textcircled{R}$ eye of patient
- $\textcircled{L}$ eye of doctor $\rightarrow$ $\textcircled{L}$ eye of patient



slit lamp microscopy : Another method to look at the fundus

Concave lens

Convex lens

HRUBY LENS
-58.6 D

non-contact

+75 D

+90 D

contact

Goldman 3 mirror
lens $\Rightarrow$ +60 D

Adv: Magnification + Stereopsis

b) Indirect Ophthalmoscopy / a.k.a Fundoscopy



- Binocular vision
- Condensing lens used $\Rightarrow$ +20 D ; +30 D
- Entire retina can be seen
- Real inverted images
- Done at arm distance from the patient
- Field of view $\Rightarrow$ 8DD area at one time
- Higher learning curve
- Not influenced by refractive error.

DIRECT OPHTHALMOSCOPE

- 1) Unocular vision
- 2) Virtual erect image
- 3) Visualise optic disc & macula
- 4) Field of view: 2 Disc Diameter
- 5) Magnification: more
- 6) Depends on R.E of pt & doctor

$$M = \frac{\text{Power of Eye}}{\text{Power of lens}} = \frac{60}{4}$$

= 15 times

Power of eye = +60 D
" of cornea = +44 D
" of lens = +16 D

INDIRECT OPHTHALMOSCOPE

- 1) Binocular vision
- 2) Real inverted image
- 3) Visualise Fundus & entire retina seen
- 4) Field of view = 8DD [1DD = 1.5mm]
- 5) Magnification: Less
- 6) Does not depend on R.E

$$M = \frac{\text{Power of eye}}{\text{Power of lens}} = \frac{60}{20 \text{ or } 30}$$

= 2-3 times

Power of Converging = Myopia > Emmetropia > Hypermetropia

(i.e. > 15 times)

eg: -4 D

$$= \frac{64}{4} = 16 \text{ times} //$$

Notespaedia.com
More than just notes

Investigation based on Purkinje Images:

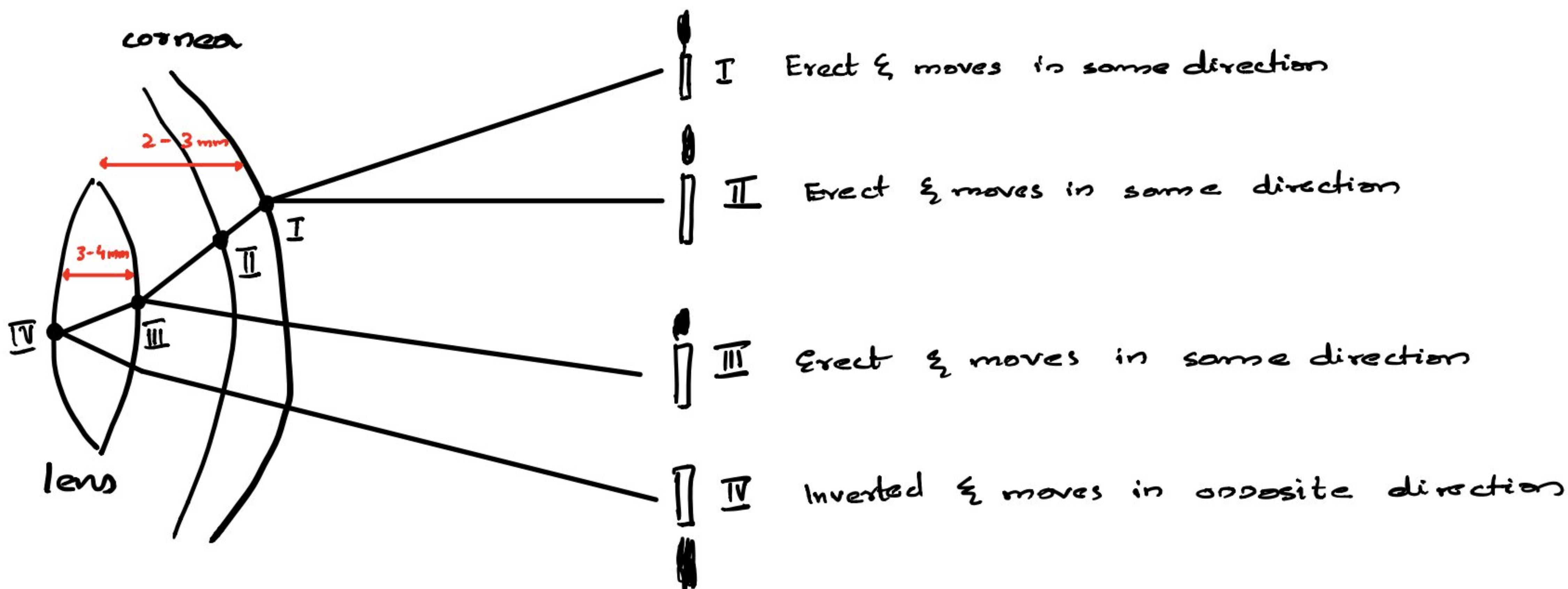
- Corneal thickness = Pachymetry $\rightarrow$ I & II
- Corneal curvature = Keratometry $\Rightarrow$ I

Q: 2/3rd of refraction of light done by cornea

Q: R.I of cornea = 1.376

R.I of air = 1.00

- Corneal surface = Placido disc $\Rightarrow$ I
- Corneal endothelium = Speculum microscopy $\Rightarrow$ II



- (N) Thickness of lens in adult = 4mm
- Ant. Chamber depth b/w II & III = 2-3mm

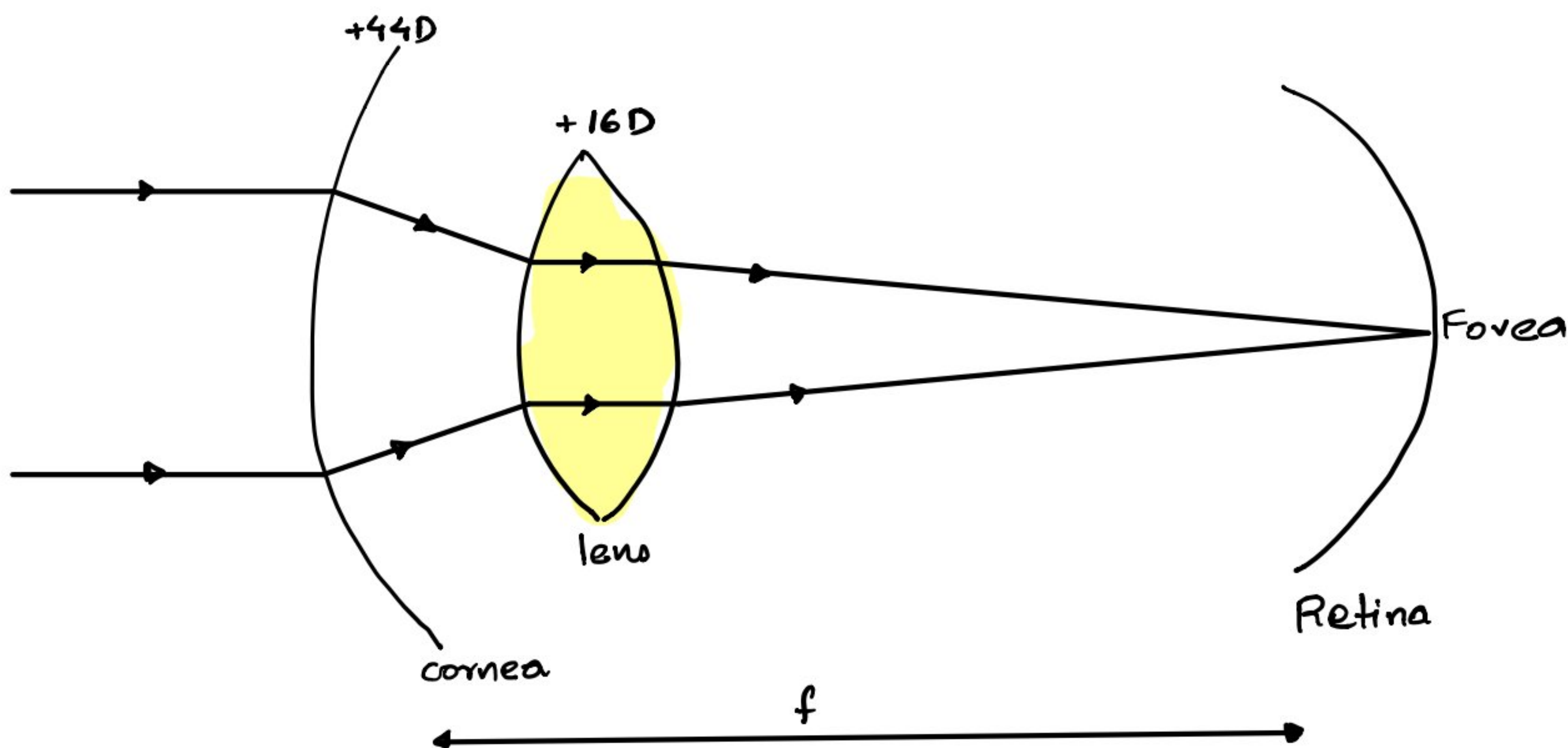
Q: Purkinje Image in Aphakia $\Rightarrow$ Image III & IV are absent
 " " • mature cataract $\Rightarrow$ Image IV Absent.

ultrasound ::

- A-scan $\Rightarrow$ tell axial length of eye
- B-scan $\Rightarrow$ post. segment evaluation in case of media opacities freq. 10 MHz

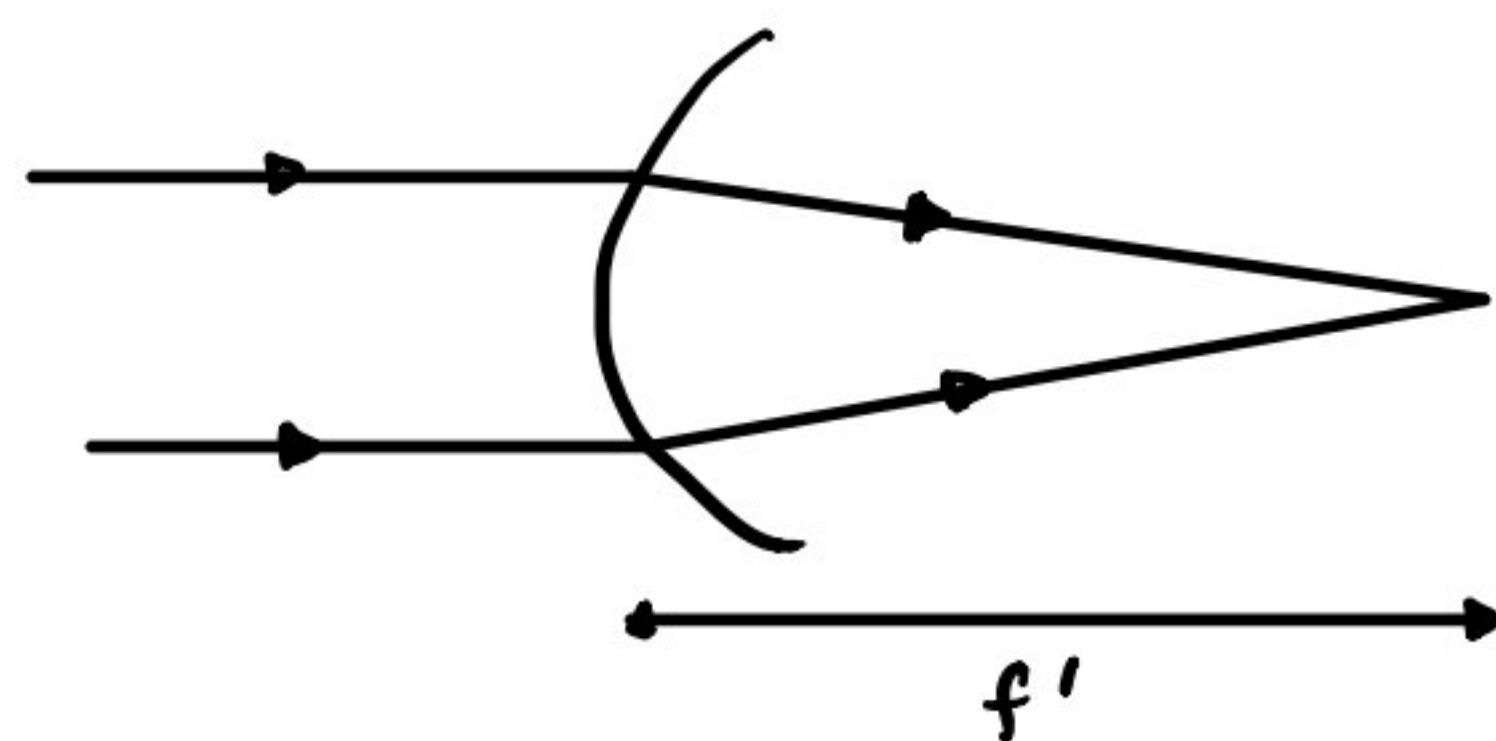
Refraction : Bending of light when light passes through two media
Refraction of eye done by cornea & lens
 $\{3/4^{th}\}$ $\{1/4^{th}\}$

Total refraction power of eye $\Rightarrow +60\text{D}$
 " " cornea $\Rightarrow +44\text{D}$
 " " lens $\Rightarrow +16\text{D}$



Refractive Power, $D \propto 1/f$ (focal length)

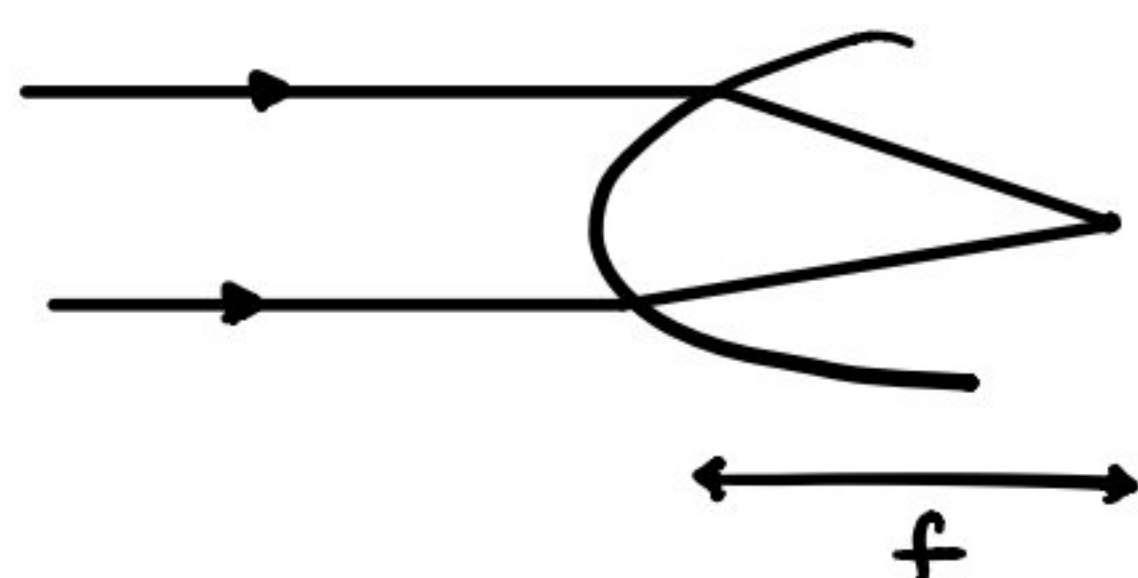
Refractive Power & Curvature:



flatter curvature

$$D \propto 1/f$$

$D \propto \text{curvature}$



steep curvature

Refractive Power vs. R.I.:

$$D \propto \mu$$

Q. Max. refractive index $\Rightarrow$ Lens nucleus
 Max. refraction $\Rightarrow$ Tear film - air interface
 $= 1.357 - 1.00$
 $= 0.357$
 $\Rightarrow$ Cornea

R.I of structure in Eye:

Tears = 1.36

Cornea = 1.376

Lens cortex = 1.386

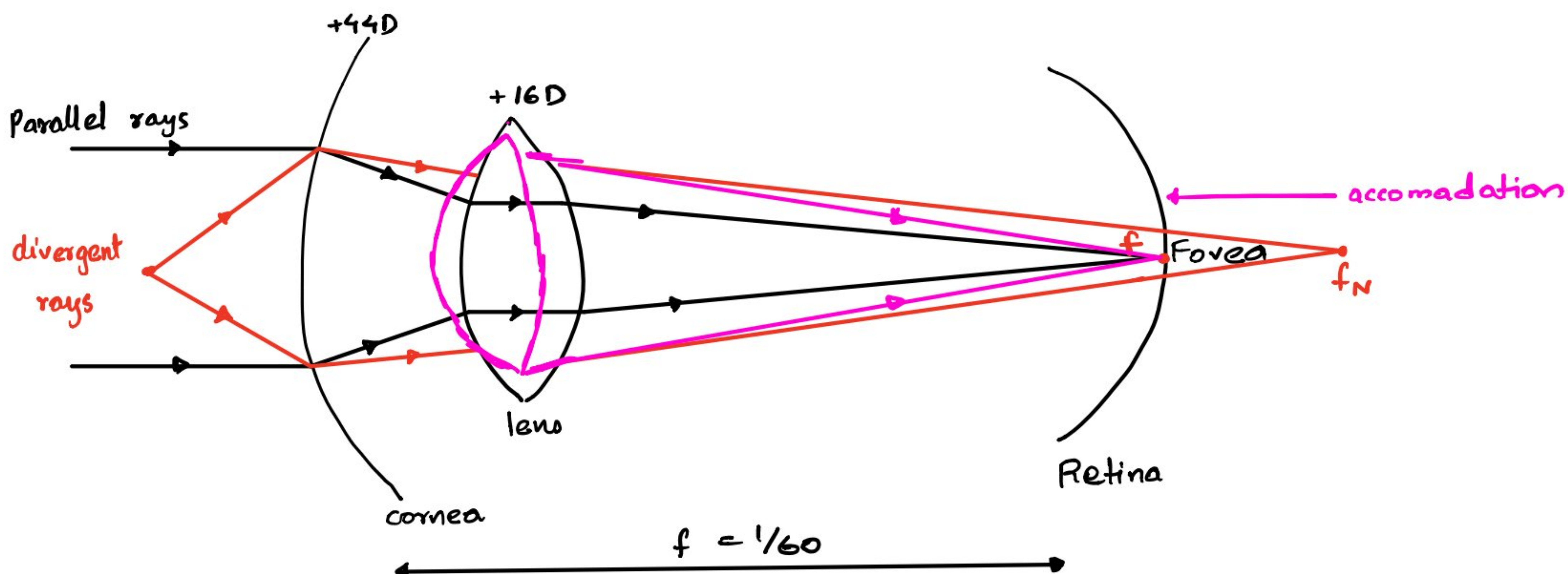
Lens nucleus = 1.406

Vitreous = 1.335

Aqueous = 1.336

Air = 1.00

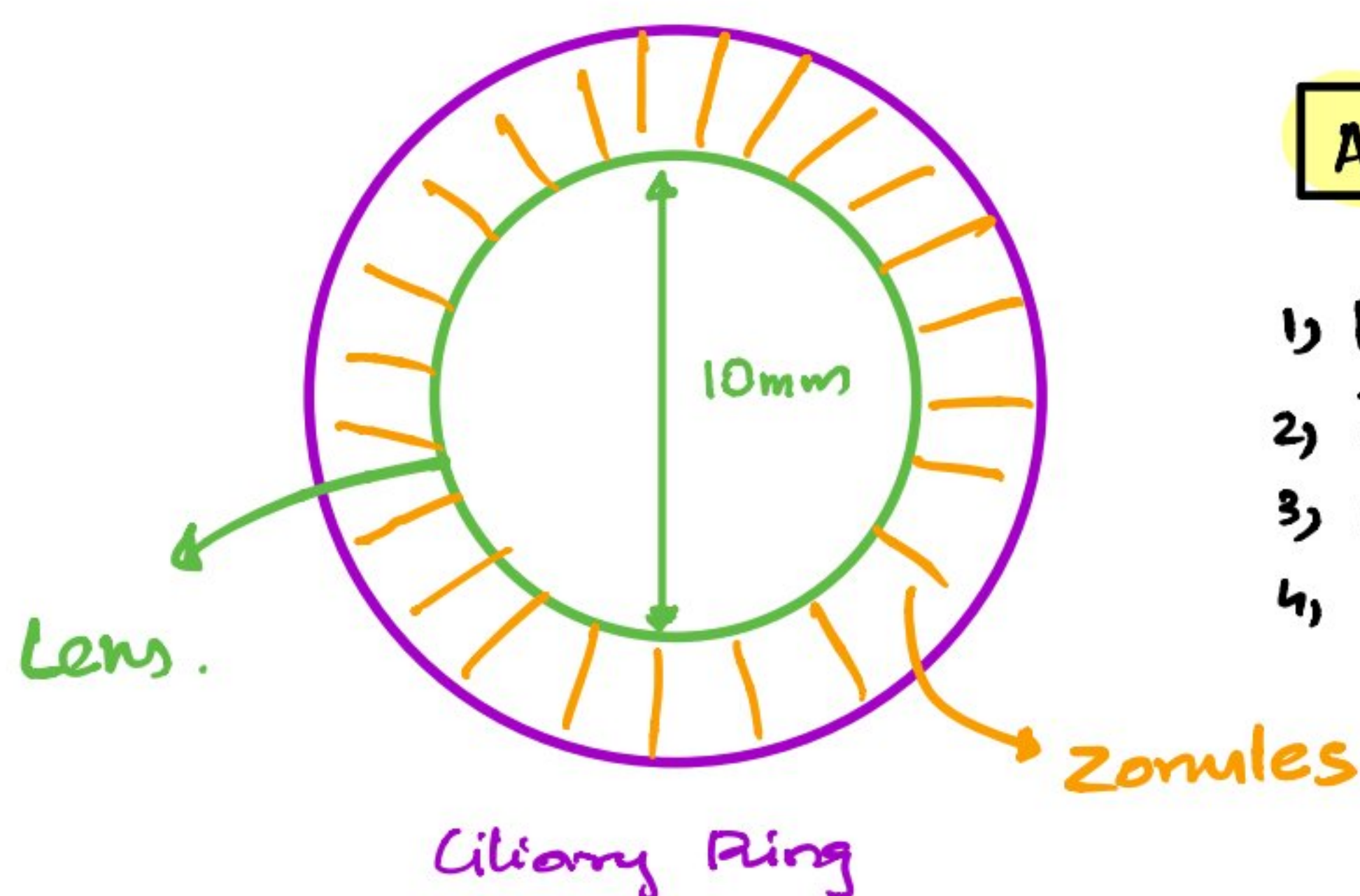
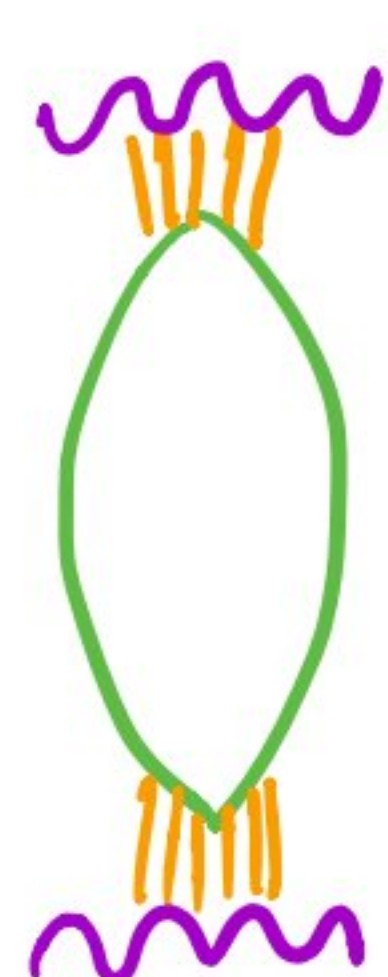
Emmetropia:



Accommodation = Process by which near objects are brought to focus on the retina by contraction of circular fibres of ciliary muscles.

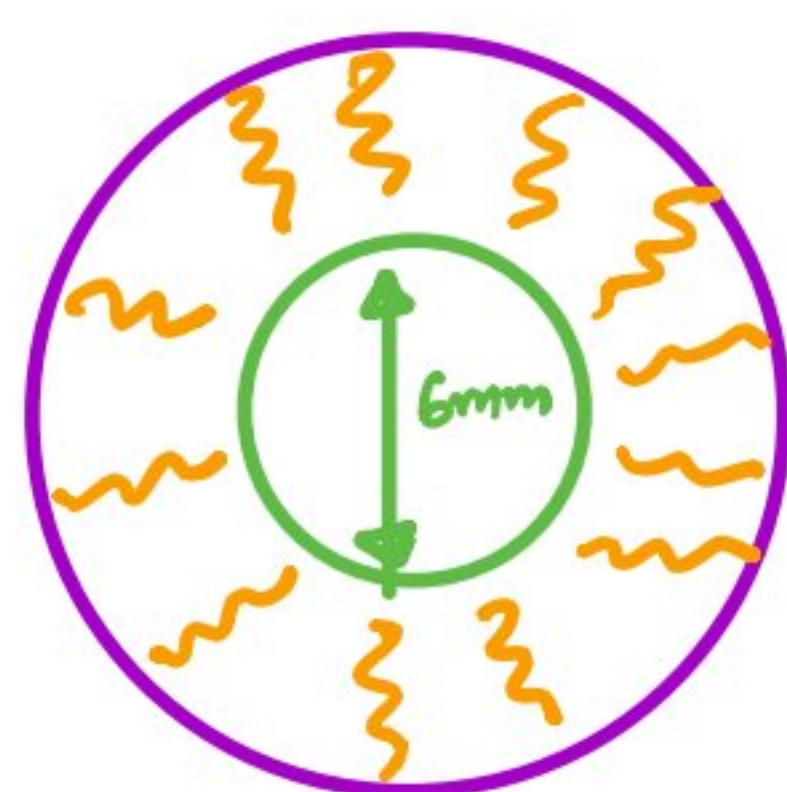
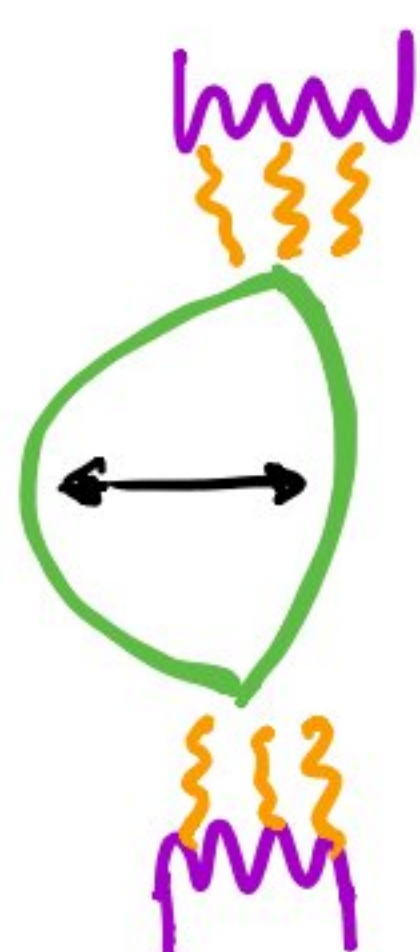
Near triad / Accommodation triad:

- 1) Convergence of both eyeballs → middle rectus → IIIrd N.
- 2) Miosis (to ↑ depth of focus) → sphincter pupillae → parasympathetic supply of 3rd N.
- 3) ↑ in Anterior curvature of lens ⇒ **Ciliary muscle** → parasympathetic supply by 3rd Nerve.



At Rest

- 1) P > A curvatures of lens
- 2) Zonules stretched & in tension
- 3) Anterior capsule is stretched
- 4) Equator diameter = 10mm.



At Accommodation

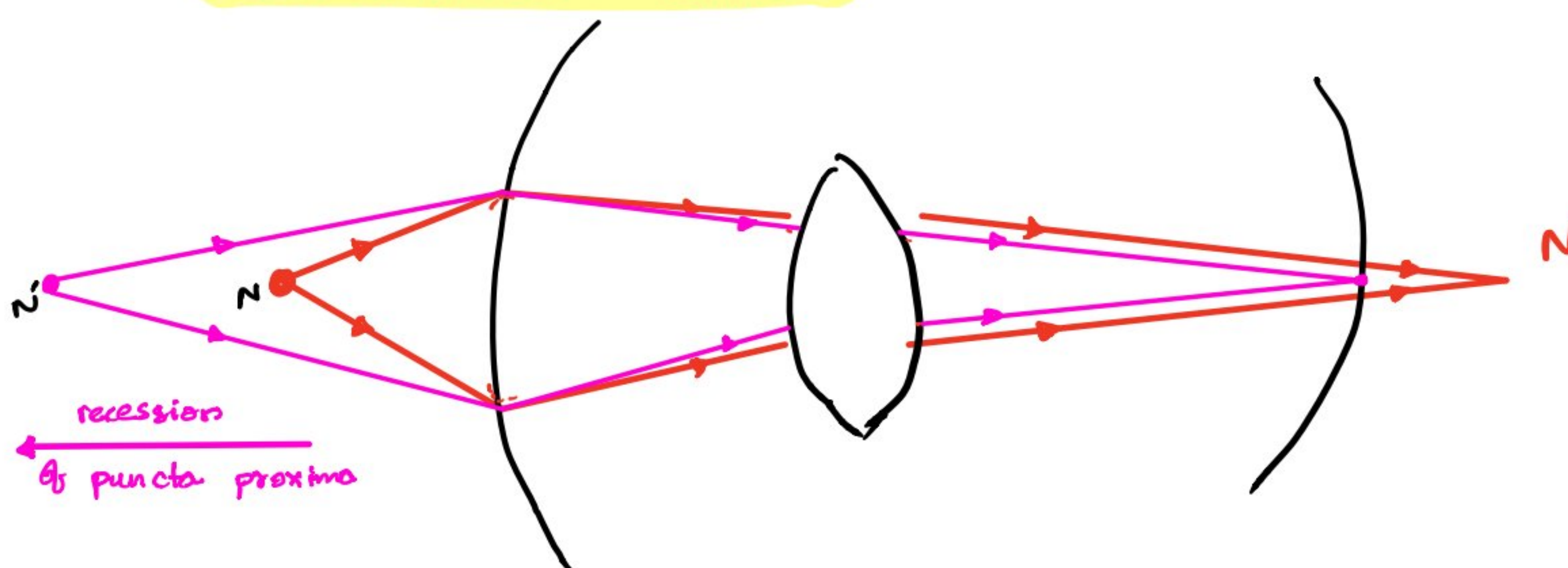
- 1) ↓ size of diameter of ciliary ring
- 2) Relaxation of Zonules
- 3) Relaxation of Anterior Capsule
- 4) ↓ size of equatorial diameter of lens
- 5) ↑ size of anterior curvature of lens
{ ∴ forward bulging of anterior capsule }
- 6) Posterior curvature of lens → same
- 7) ↑ Dioptric power of lens
- 8) Thickness of lens ↑
- 9) More spherical lens

For Accommodation :

- 1) Ciliary muscle tone
- 2) Elastic anterior capsule
- 3) Lens substance → mouldable

↓ Old age

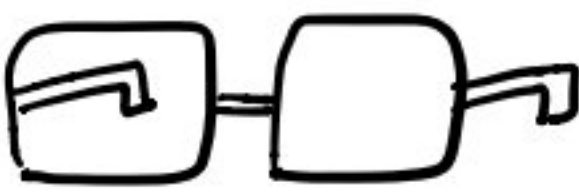
① **PRESBYOPIA** : Physiological loss of accommodation d/t ageing
≠ Not Refractive Error



Result in
Recession of
puncta proxima
(near point)

Correction :: $\uparrow D \longrightarrow \downarrow f$

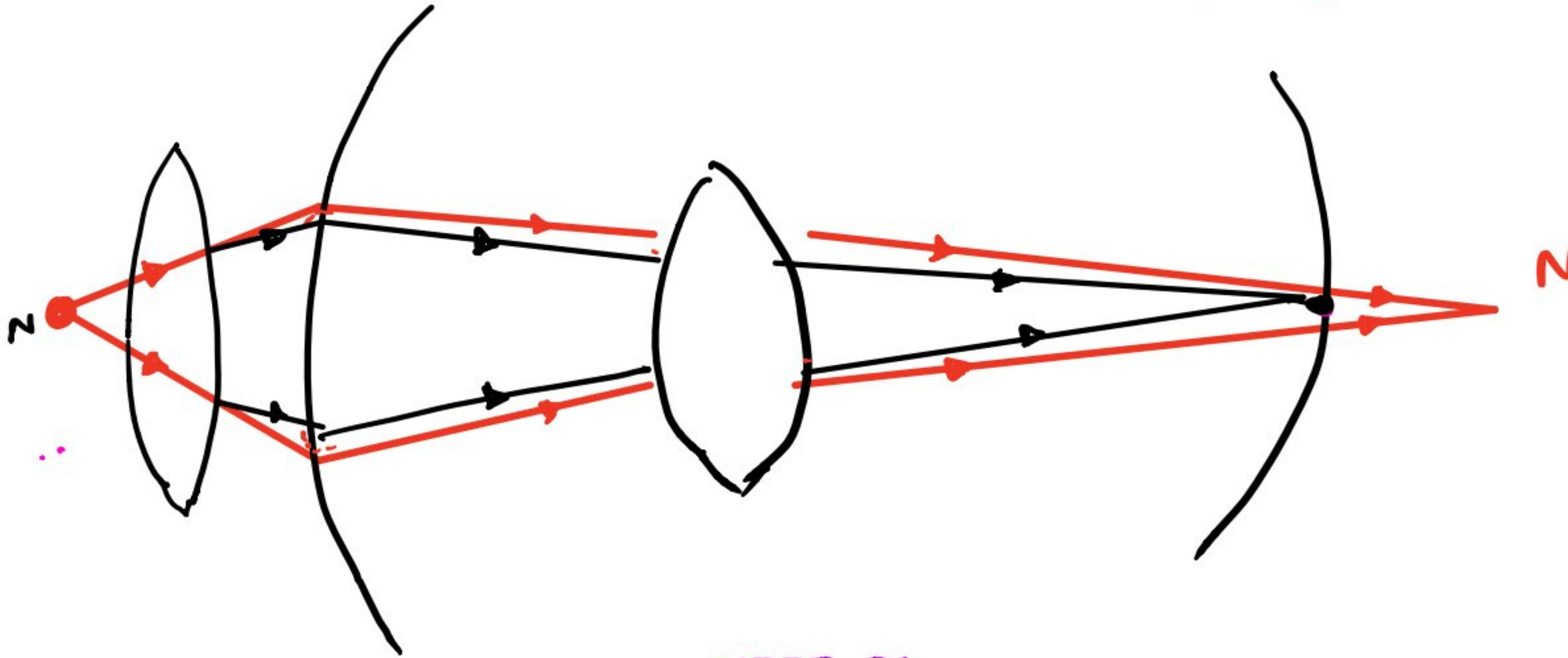
① Convex lens
or positive spherical lens

② Progressive lens: 

③ Executive Bifocal: 

④ Kryptok Bifocal: 

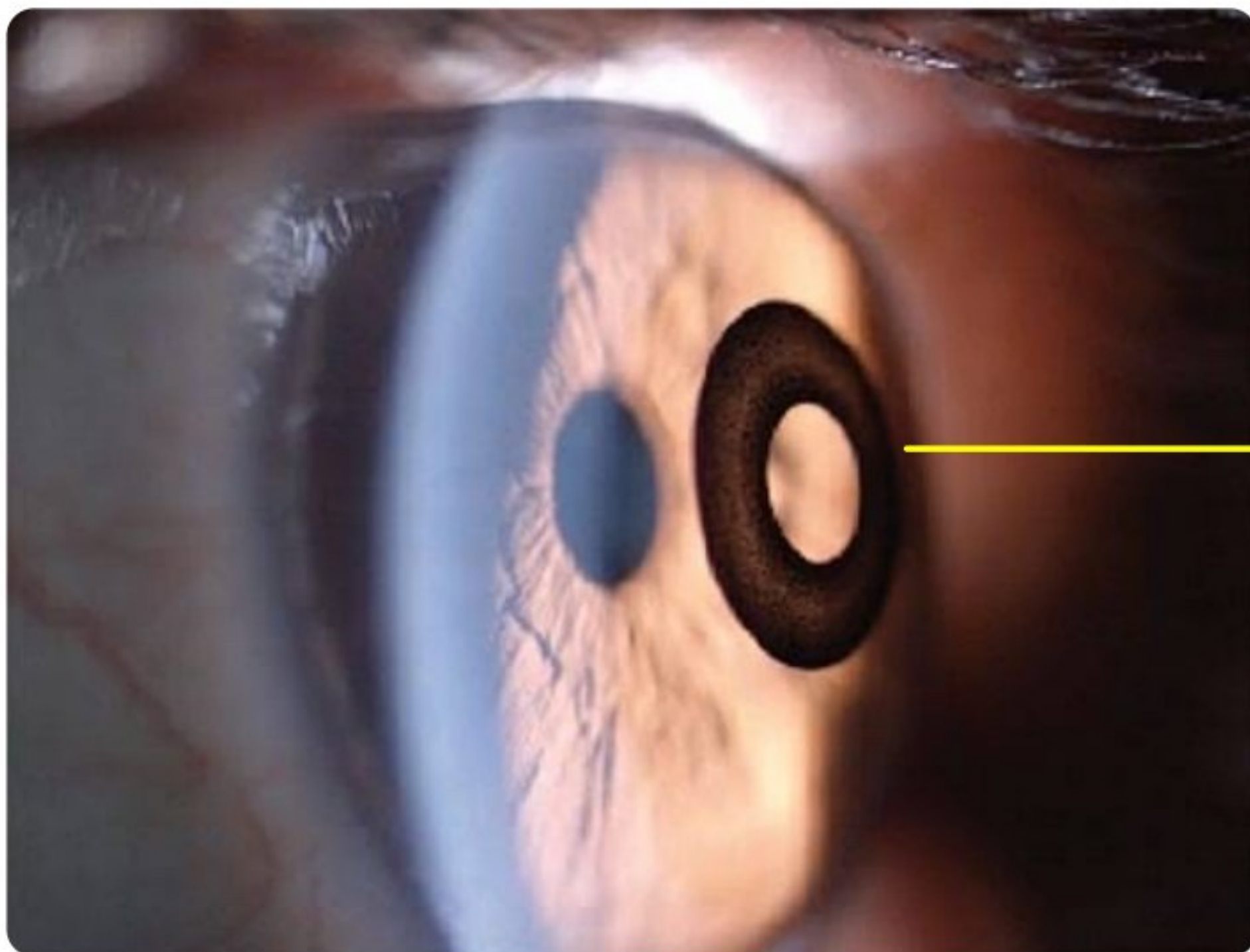
⑤ Tri-focal: 



NEET-PG

Age	Power
40-45 yrs	+1D
45-50 yrs	+1.5D
50-55 yr	+2D
55-60 y	+2.5D

② KARMA INLAY



Kamra Inlay

Pinhole Mechanism

- ↓ aperture
- Eliminate Peripheral ring
- ↓ spherical aberration
- ↑ depth of focus

③ Cataract Patient:

IOL $\longrightarrow$ multi-focal IOL for all distant intermediate near } vision